Supplementary Material for "Learned Reliability-Aware Fusion of Token Evidence Raised Image-Text Verification Accuracy Under Gaussian Noise in a Unified Masked Diffusion Model: A Preregistered Confirmation in UniDisc"

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References to sections, tables, equations, and propositions of the main paper are written in words (for example "Section IV-C of the main paper"); labels with an S belong to this supplement. Every number carries a source comment in the LaTeX file, with its `runs/` file and key. Study 1 material (S-II, S-III) and Study 2 material (S-IV, S-VI, S-VII) are kept in separate sections and separate tables.

S-I. Preregistration Record

A. Pins

Study 1's artefacts are pinned in Table S1 and Study 2's in Table S2. Hashes are SHA-256 of the committed bytes (git LF convention) unless marked. Some working-tree files are CRLF, so a raw hash of a checked-out file can differ from the committed hash without a change of content; both forms are recorded where both were used.

TABLE S1
Pinned Artefacts of Study 1 (pin file `runs/P2-GATE-CARD-V4/pins.json`)

Artefact	Identifier
Code, tests, guard	final audited commit `9c4e1e25`; moved by dated rows A-1 (`f2f7b2e1`), A-4 (`d81858c0`, Tier-3 code added), A-5 (`8d1bee3a`, re-infilling self-check), and A-6 (`725a1b5a`, family H and Study 2 code); manifest hash `65513677ba4d98adfa1a861db4215200ad64b78aec1cceeaf0bcdca6b4664dec`
Analysis plan with Addenda A1 to A3b.12	`e33f078c45b92b94cbf658192e2fd815f8c6eafbffe53ba39bc0dd728982afb7` (LF)
Addendum A4 (outcome book, Tier-3 comparators)	draft commit `40828e96`, revision r1 commit `f1671c91`
Addendum A5 (family H)	final `b32e72a1`, revision r1 `2f62f1df`; family H code commit `5e6380b1`
Fold file	LF `c19db5b3b9fabdeb89e8e686c58ae6c011f746529fbf9be84f1c58c99c446786`; CRLF working copy `450edb543779de7ee5e6a52422fd6507ef52f15ba3b956a8b05c2e561620f311`
Hard-negative pool (family B), manifest, leak check	`cfd0f94facd0`, `070fb29a2ebc`, `91f4b7968b54` (prefixes, LF)
Frozen files of E-pred	fit `c89700dbc3ba`, signal-to-noise calculator `5c535a3bda75`, test file `8d7528bbb37b`

TABLE S2
Hash Set of the Frozen Weighting of Study 2, Verified by the Scorer at Start

Artefact	SHA-256
Model `runs/P2-LRF-FROZEN-LFIX/model.npz`	`db7fdb4e9f863a18148a4567d396c36a1bba58dc8b041f079492b10466770f37`
Specification `spec.json`	`9d196d7189dc5b6a2d6d80ed694314c6ed323cac59b0300c42c073a6e40db4b5`
Tuning record `hpo.json`	`63cd79442078414726b327eef76e1e1397b3db1b7c3222449705a3f4acdad04f`
Feature code `data.py`	`14421ffb1061947a8408406750e8417282188efeccf07cac3142179fed9f429b`
Model-family code `families.py`	`1c49827d723ea17017f84ba283fcfa681bf24e10c0a0e7d62b6a83064f850a7e`
Fitting code `lrf.py`	`03ed472e1a3cd8bb2dc41f4a59d7b48f1561cd883012ac4434e08a5c37b9d6ec`
Codebook (untracked)	`ff15b9b5b362dc80f8a18f6e852e6e99478c0e8885646735ed16d1d80e3ae50b`
Cached position marginal at slot length 13 (untracked)	`f7744253a1b997b8b9f28d9b91348e8371f1ea1e3564f6c8bfc2f10aa5b42176`
Re-scored design arrays (digest)	`038ded0724b33e4ff132ce9af4eefd7c115bf82fc834777be1e2f4d69cb68e90`
Fresh fold `folds_lrf.json`	CRLF working copy `a545a83ec881988cc81d40c3598fcb4be6292a59d2c835e4cce6e72d3d71f1de`; the committed LF blob has another hash for the same content

B. Audits

The checks named in the main paper as automated audits run in the authors' own pipeline (not independent reviews) are: the plan review (sign-off of the analysis plans); the adversarial review (signatures with conditions on Addenda A2 to A6 and on the build specification); the code, commit-order, and hash audit (pass verdicts at the final audited commits, with delta audits for every later pin; gate audits of family L, family LG, and the reference encoders with their parity runs; byte-identical regeneration of the committed outputs after every code merge); and the fold and leak checks (sign-offs of the hard-negative pool and of the Study 2 fold, check L7). These checks were run by the authors' own pipeline; none is described as independent. No signature says that a method works.

C. Families

Table S3 lists every test family of the paper. Each family has its own level 0.05 and its own Holm correction; there is no error-rate control across families. The paper has 17 families; under a global null with independence the chance of at least one false rejection is $1 - 0.95^{17} = 0.58$.

TABLE S3
TEST FAMILIES. m_{H}: HYPOTHESES IN THE FAMILY'S HOLM CORRECTION

Study	Family	n	Test	m_{H}	Not allowed
1	P	300, 500, or 300 with the evidence ratio	exact two-sided McNemar, GATE-M against none: JPEG, noise, splice	3	a claim in favour of the gate from a rejection with $c > b$; "predicted", "explained", "caused by detection"
1	PS	as P	same test on patch and swap	2	use as a demonstration that the gate works
1	S2	300	exact McNemar, re-infilled flagged set against random-gate best-of-four re-infill (kept draw), patch and splice; only if $n \geq 120$	2	"improves", "restores", detection as mechanism
1	B	300 pairs	exact McNemar, two-way fusion against the better single direction; untestable at ceiling 0.95	1	a statement about corruption
1	R	150	exact McNemar per corruption with a primary rejection with $b > c$ (none: not run)	–	any claim beyond the tested model
1	S-M, S-Z, S-DN, S-D, S-D2, S-C	as P	mechanism, evidence ratio, denoising, detection, out-of-sample detection, clean control	3, 3, 2, 2, 2, 1	as in the protocol
1	H, H-M	300 (planned)	as P on SugarCrepe hard negatives; added after the primary result	3	see S-III
2	L	450	L1 exact one-sided McNemar on noise; then Holm over L3 (JPEG), L4 (splice), two-sided	3	"confirmed" except for L1
2	L-NI	450	lower 90% bootstrap bound of the clean net change against -0.03	1	a superiority claim on clean images
2	L-LEN	449	exact one-sided McNemar on noise with length-matched negatives	1	–
2	LG	300	exact one-sided McNemar, shot, impulse, speckle (severity 3)	3	a claim beyond the three tested noise types
1	E-dec, E-pred, suite	P fold	intervals or sign agreement only; no test	–	any p-value in a claim

D. Dated Amendments

Row A-1 moved the Study 1 code pin for later stages; it changed two command-line flags of the stage runner and the guard settings recorded per chunk, and no analysis path, scoring definition, or frozen file. Row A-2 set a slow-item limit of three times the median item time, with a floor of 2 s, for later stages on the first author's dated answer; guard settings decide only whether a run continues. Row A-3 closed the window for tightening the second-model stage before the main analysis ran; that stage never ran. Row A-4 added the audited Tier-3 code, with no existing file changed. Row A-5 made the unchanged tokenisation self-check of the re-infilling stage run with deterministic convolution algorithms. Row A-6 merged the family H and Study 2 code after the re-infilling stage completed, with a byte-identical regeneration of every committed Study 1 output. Row A-7 recorded the family H preconditions before its first item. Row A-8 made the Study 2 self-check run with deterministic convolution algorithms after two stops before any item (pin `1871fe7d`). Row A-9 changed one test only (pin `84be25c4`); production code was byte-identical.

E. Time Input of the Released Configuration

The main paper (Section III) states that UniDisc's denoiser has no time input in the released configuration. Table S4 gives the evidence, read in the authors' code at commit `3ee3e47a9e64bac2fdca59957726f14723371ced` of the UniDisc

repository and in the key names of the checkpoint snapshot `2b344ca856bd108b52a2d022a3463dace131249f` (a code reading, not a reproduction). The statement concerns this configuration and checkpoint, not UniDisc in general. Because the loader is non-strict, a key check was run: all 342 checkpoint tensors match the 342 parameters of the denoiser backbone, with no missing key, no unexpected key, and no shape mismatch. The check does not cover the image tokeniser, which is loaded by another path, and key equality does not show that the values are the intended final weights; how often whole-modality masks occurred in training cannot be recovered from the code.

TABLE S4
Evidence for the Absence of a Time Input (Paths Relative to the UniDisc Repository Unless Marked)

Claim	Evidence (file:line)
Time conditioning is off; absorbing-state diffusion, parameterisation `subs`; no experiment overrides it	`configs/config.yaml:12,14,15`
The project composes the official demonstration experiments and loads the interleaved checkpoint	project files `runs/env-2026-10-14/scripts/unidisc_setup.py:17,30-35`, `exp001/unidisc_net.py:46-47`
The official inference experiment sets a null noise level; the samplers pass `None`	`configs/experiments/large_scale_high_res_interleaved_inference.yaml:30-31`; `model_eval.py:2057,2086,3019,3060,3186`
The forward pass accepts a null noise level only when this is allowed	`model.py:660-662,691,729`
No timestep embedder and no time modulation without time conditioning	`models/dit.py:1105,1186-1188,1377-1379`, `models/dit.py:1064-1090`
A stored scaling flag is never read	`configs/model/extra_large.yaml:8`; `models/dit.py:1290`
No time-embedding or time-modulation keys; "sigma" absent from the checkpoint	key names read from the checkpoint pickle
Raw logits are returned before the step that copies observed positions	`model.py:784-788`

F. Protocol Details of Study 1 (Main Paper, Sections IV-A and IV-B)

Each realistic corruption has two rungs, walked in order: JPEG quality 10, then 5; Gaussian noise with $\sigma = 0.10$, then 0.20; a pixel splice of 128, then 192 px. A rung passes if the size-normalised evidence $\bar{\Delta}/s$ falls under corruption and recovers under the changed-code gate, which keeps exactly the tokens whose code did not change (paired $t \geq 2.0$ for both, $n_v \geq 90$ items). The first passing rung is the operating point. The largest headroom discordance D^* between the changed-code gate and no gate then sets the test size: 300 items if $D^* \geq 29$ (rule R1), 500 items if $18 \leq D^* < 29$ (R2), and a switch of endpoint if $D^* < 18$ (R3). The design minimum detectable net gain at $n = 300$ was 0.053 unadjusted, under an assumed discordance of 10%. Ties of the suspicion scores from `bf16` logits are broken by an independent uniform jitter smaller than the smallest gap between distinct calibration scores. Subtracting D_j gives the Neyman–Pearson statistic only for corrupted values drawn from the position marginal, which JPEG compression, noise, and splices do not produce, so no statement about detection power is made.

G. Disclosures of Addendum A4

(1) The five Tier-3 comparators and their parameters (S-II) were fixed in Addendum A4, whose draft (`40828e96`) and revision r1 (`f1671c91`) were committed before the headroom decision file of the main run was written; the addendum's author states that no value of the headroom or primary stage was read when it was written, which is an assertion and not an attestation. The reading rules for few discordant items (revisions r2 and r3 of the addendum) were added after the headroom decision file was committed and read, before any main-test value or Tier-3 output existed. (2) No p-value is computed for the Tier-3 comparators; there are 72 intervals and no multiplicity adjustment. (3) The outcome book of Addendum A4 fixed the wording of every label before the primary result existed. On 28 October 2026 at about 18:40 UTC the first author delegated the choice of how the submission condition reads under a null primary result; under that delegation the stricter reading (no question-form framing in the title, abstract, highlights, conclusion, or the opening of the Study 1 discussion) was adopted within the authors' study pipeline at 18:42 UTC, before any value of the main test was read. Decision records D0076 and D0077 document this; D0077 corrects D0076's wording (the author's message delegated the choice and did not itself choose a reading). (4) The time-input statement rests on a code reading at the pinned commit (above) and on the key check. (5) Family B ran before the re-infilling family S2, which the plan allows when a fixed time inequality holds at the end of the coherence stage (the current time plus the planned duration of the re-infilling stage and fixed margins had to fall before a deadline fixed in Addendum A4); it held. A code reading showed that no output of family B can reach an input of S2; the rule that B is analysed only after the S2 table exists is procedural, and it was followed (analysis commit `10e6bc0f`).

H. Full List of Deviations and Disclosures

Study 1.

1) The sample size of S2 and B was set to 300 instead of 150 before any outcome, on the first author's dated confirmation; S2 was restricted to patch and splice after a 21-item dry run.
2) The first stage of the detection measurement stopped and was resumed at item 106 of 200.
3) Three checks of the second-model adapter are applied outside the module that holds the others and were reviewed, not executed, by the auditor; the second-model stage never ran.
4) The sensitivity analyses of S2 and B read "rejection" as the favourable-direction rejection at the adjusted level; this can over-flag sensitivity to n, never under-flag it, and it enters no claim.
5) The firewall for the S2 smoke test is an honour rule.
6) The swap and patch parameters are stored in one frozen file that stands for two files named in the plan.
7) The chance bar for the sanity corruptions is called blind on the lead analyst's assertion that no detection-pilot output from item 26 on was read; the first author's written co-attestation of not having looked at those outputs is dated before the analysis ran.
8) Where the card forbids any claim from a rejection with $c > b$ and the plan prescribes "reduced verification accuracy", the plan governs, and no claim in favour of the gate is made from such a rejection.
9) The calibration stage stopped twice on the slow-item rule of the guard (106 and then 114 rows complete), with no crash; it was relaunched on the first author's dated answers (row A-2); stored rows were kept unchanged.
10) The main-test stage stopped once at 334 of 500 on the default chunk limit of the stage runner, not a guard stop, and resumed at row 334 with the same items and flags.
11) During the coherence stage the workstation restarted for an update after one of 100 items; the stage was continued on the first author's instruction and completed 100 of 100.
12) Family B ran before S2 under the plan's time inequality (A4 disclosure (5) above).
13) No second-model stage ran, because no corruption had a primary rejection with $b > c$.
14) The S2 smoke test stopped twice at the tokenisation self-check before any item, caused by process-to-process nondeterminism of autotuned convolutions in the image tokeniser; row A-5 fixed the self-check, and S2 reads stored codes and never re-tokenises.
15) Family H: the A5 disclosure (S-III) and amendment items (A5.14, A5.1 amendment item 7); decision D0087 was committed after the H3 driver had started and before any H3 row existed.
16) The effective-n record of the S2 figure carries a reason text written at analysis time.
17) The fold builder's notes: the image ids of the negatives and mismatches were not collected separately (no analysis uses them); 162 other `runs/P2-*` JSON files outside the excluded classes were not opened by the fold check.
18) The time-input table is reported in this supplement only.
19) The exact Clopper–Pearson bounds of the family P table were added at writing (`runs/P2-PAPER-BOUNDS`, descriptive).

Study 2.

1) The L1 stage stopped twice at its tokenisation self-check before any item (directories `P2-LRF-L1.try1`, `.try2`); row A-8 moved the pin to `1871fe7d`. Residual: the self-check certifies the layout under the deterministic backend, while the items ran with autotuning on as in the design run, and a fresh tokenisation can differ by up to about 1.5% of tokens between processes.
2) The scorer's pin file was written at 22:42 while L1 was running; every value in it was copied from records committed before L1 (the hash set of the object commit).
3) Row A-9 (pin `84be25c4`) changed one test only.
4) The environment lock files of the design-run, re-score, L1, and LG stages are empty; the environment was recorded after the runs (`runs/P2-ENV/pip-freeze-2026-10-30.txt`); the commit of the corruption package was verified from the environment.
5) Item arrays are untracked; SHA-256 manifests are committed.
6) Hashes appear in CRLF and LF forms; both refer to the same content.
7) The parity files were moved unchanged from a wrong output directory; no hash was taken before the move, and audit-time SHA-256s are recorded in the reference-encoder audit.
8) The field `pixel_parity: not run` of the main reference-encoder file is superseded by the two parity runs (S-VI).
9) The download of the reference encoders was 5.2 GB against 2 to 4 GB quoted to the first author.
10) Decision D0094: what had been seen before the re-freeze (main paper, Section V-A).
11) "No output was read before its run was complete" rests on the operations log.
12) The Tier-3 ablation (S-VII) was added at writing; no ablation was preregistered.

S-II. Study 1: Secondary and Descriptive Families, and the Tier-3 Comparison

A. Family P: The Full Preregistered Test

Table S5 gives the full test of family P; its intervals are plotted in Fig. 4 of the main paper.

TABLE S5
Family P: GATE-M Against No Gate on the 500 Test Items

Cell	n	Acc. none	Acc. GATE-M	b	c	n_d	Net diff.	p (p_{Holm})	Preregistered decision
P1: JPEG, quality 10	500	0.838	0.836	0	1	1	−0.002	1.00 (1.00)	not rejected; 90% interval inside (−0.05, 0.05)
intervals: 90% [−0.006, 0.000]; 95% [−0.006, 0.000]; 98.33% [−0.008, 0.000]; exact 95% bound on |net change| 0.011 (added at writing, descriptive)									
P2: noise, $\sigma = 0.1$	500	0.676	0.676	0	0	0	0.000	1.00 (1.00)	not rejected; 90% interval inside (−0.05, 0.05)
intervals: 90% [0.000, 0.000]; 95% [0.000, 0.000]; 98.33% [0.000, 0.000]; exact 95% bound on |net change| 0.007 (added at writing, descriptive)									
P3: splice, 128 px	500	0.876	0.872	1	3	4	−0.004	0.625 (1.00)	not rejected; 90% interval inside (−0.05, 0.05)
intervals: 90% [−0.010, 0.002]; 95% [−0.012, 0.004]; 98.33% [−0.014, 0.004]; exact 95% bound on |net change| 0.020 (added at writing, descriptive)									

The main paper shows the P-only summary figure; this table holds the full family P record, including the 98.33% interval. b: items that GATE-M answers correctly and the ungated rule incorrectly; c: the reverse; $n_d = b + c$; net difference $= (b - c)/n$. Paired percentile bootstrap intervals (10 000 resamples) at 90% (the level of the preregistered decision), 95% and 98.33% (the Holm level of the smallest-p step); zero width when $n_d = 0$. p: exact two-sided McNemar; p_{Holm}: Holm over the three corruptions. The exact Clopper–Pearson 95% bound on the absolute net change was added at writing and is descriptive, not preregistered. No test rejected, so nothing is bold.

Table S6 gives the status of every Study 1 family, with no estimate and no interval (an earlier overview with estimates was deleted; decisions D0096 and D0102). Each family's estimates are in its own table or paragraph (family P in Table S5; the others below and in S-III). None of the material in this section enters a statement about family P.

TABLE S6
Study 1 Family Status Map (Status Only; No Estimates)

Family	Contrast	Status
P	GATE-M against no gate: JPEG, noise, splice	run, formal
S-M	GATE-M against the geometry-matched random gate	not run (gatekept: P+ empty)
S-Z	size-scaled evidence ratio, secondary endpoint	run, formal
S-DN	GATE-M against the denoised copy: JPEG, noise	run, formal
S-D	detection of changed tokens against cheap scores (detection items)	run, formal
S-D2	the same contrast on the test items: JPEG, noise	run, formal
S-C	GATE-M against no gate on clean images	run, formal
PS	positive control: patch, swap	run, formal
S2	re-infill of the flagged set against random-gate best-of-4 re-infill	run, formal
B	one-word foil pairs: combination of both directions against the better single one	run, formal
R	GATE-M against no gate on the second backbone	not run (gatekept: P+ empty)
E	sign agreement of the withholding theory	withdrawn (AB.3)
E-pred	out-of-sample sign agreement, five cells	run, descriptive
E-dec	three-part split of the accuracy lost to corruption	run, descriptive
Suite, Tier 3	other corruptions and detector-free combiners	run, exploratory
H	GATE-M against no gate on hard-negative pairs	headroom not met (H-NOGO); H4 not run
H-M	GATE-M against the geometry-matched gate on the H corruptions	not run (gatekept: H empty)

Status words only, as fixed by the plan; no estimate, interval, or p-value. The estimates of each family are in its own table or paragraph.

B. Headroom

Table S7 gives the headroom check that fixed the operating points and the test size. On the size-scaled ratio z_R, a sensitivity analysis, the second condition fails at every rung (T_2 from −6.75 to −0.09); the rule uses the size-normalised ratio.

C. Calibration and Flags

In 2 000 random re-splits of calibration and test items at $\alpha = 0.05$ (exploratory), the mean of $\hat{F} - \alpha$ was −0.0031, with 2.5 and 97.5 percentiles of −0.0055 and −0.0006. The clean no-gate accuracy on the test fold was 0.880 (Wilson 95% interval 0.849 to 0.906), against 0.843 (0.798 to 0.880) in the pilot; the intervals overlap, so the pre-stated pipeline audit was not triggered. For $\alpha = 0.05$ on an 8×8 patch image about 48 false flags are expected against at most 64 true ones, so precision is at most 0.57 even for a perfect detector, and under JPEG and noise more than nine in ten tokens change, so precision must be read against that base rate.

D. Positive Control (PS, Pipeline Check)

Table S10 is the positive-control table of Section IV-C of the main paper. It is a pipeline check and is not read beside family P.

TABLE S7
HEADROOM CHECK ON THE 100 HEADROOM ITEMS (NO GATE FLAG COMPUTED)

Rung	n_v	T_1	T_2	Passes	D_c	Changed codes	Accuracy drop	Operating point
JPEG, quality 10	99	6.03	3.30	yes	19	986.7	0.06	yes
JPEG, quality 5	95	5.16	1.54	no	26	1010.5	0.04	–
Noise, $\sigma = 0.10$	100	4.20	4.38	yes	18	949.4	0.14	yes
Noise, $\sigma = 0.20$	100	3.79	4.27	yes	19	998.4	0.22	–
Splice, 128 px	100	5.33	4.05	yes	2	237.2	0.01	yes
Splice, 192 px	100	7.93	7.34	yes	3	387.7	0.03	–

n_v: items with all values; T_1, T_2: paired t statistics of clean minus corrupted and changed-code gate minus corrupted, on d_R; a rung passes if $n_v \geq 90$, $T_1 \geq 2$, and $T_2 \geq 2$, and the first passing rung is the operating point. D_c: items whose correctness differs between the changed-code gate and no gate. Changed codes: mean number of the 1 024 codes that differ from the clean tokenisation. Accuracy drop: clean minus corrupted accuracy without a gate. Rungs after a passing first rung were computed but not used. No interval is reported, so nothing is bold. $D^* = \max(19, 18, 2) = 19$ over the operating points gives rule R2 ($18 \leq D^* < 29$): 500 test items, endpoint Y. Source: runs/P2-GATE-M3/family.json, keys rungs.* and n_rule.*.

TABLE S8
CLEAN FALSE-ALARM CHECK ON THE 500 CLEAN TEST PAIRS

α	$\hat{F}_{\text{cal}}$	$\hat{F}$ (test) [95%]	Z_{cal}	Reading
0.01	0.0067	0.0064 [0.0061, 0.0067]	−16.12	no evidence of violation
0.05	0.0468	0.0456 [0.0442, 0.0470]	−3.59	no evidence of violation
0.10	0.0970	0.0956 [0.0928, 0.0985]	−1.76	no evidence of violation

$\hat{F}$: mean over items of the flagged share of an item's 1 024 tokens; interval: percentile bootstrap 95% over items (10 000 resamples). $\hat{F}_{\text{cal}}$: flagged share on the 300 calibration pairs at the fitted threshold. Z_{cal}: calibration statistic of Section S-I; a violation is flagged above 2.13. Nothing is bold (no interval of a difference). Source: runs/P2-GATE-M4/analysis.json, keys calibration.fap_check.per_alpha.*; runs/P2-GATE-M2/threshold.json, keys thresholds.*.r_hat.

TABLE S9
FLAGS OF THE GATE AT $\alpha = 0.05$ ON THE CORRUPTED TEST IMAGES (TOKEN LEVEL, POOLED OVER THE 500 ITEMS)

Corruption	Changed share	Hit rate	False-alarm proportion	Precision
JPEG, quality 10	0.961	0.061	0.037	0.976
Noise, $\sigma = 0.10$	0.928	0.033	0.036	0.922
Splice, 128 px	0.227	0.049	0.044	0.248
Patch (sanity)	0.062	0.750	0.050	0.498
Swap (sanity)	0.150	0.859	0.089	0.631

Changed share: share of tokens whose code differs from the clean tokenisation (ρ). Hit rate: share of changed tokens flagged. False-alarm proportion: share of unchanged tokens of corrupted items flagged (not covered by the guarantee). Precision: share of flagged tokens that changed. Point estimates; no interval was computed, so nothing is bold. Patch and swap are the sanity corruptions of the positive control. Source: runs/P2-GATE-M4/analysis.json, keys calibration.per_condition.*.{precision,recall,fap_on_clean_tokens} and calibration.per_condition.*.prop4.rho.

TABLE S10
POSITIVE CONTROL (PIPELINE CHECK): GATE-M AGAINST NO GATE ON THE SANITY CORRUPTIONS

Cell	n	Acc. none	Acc. GATE-M	b	c	n_d	Net diff.	p (p_{Holm})	Preregistered decision
PS1: patch, 8 × 8 codes	500	0.870	0.870	2	2	4	0.000	1.00 (1.00)	not rejected; positive control not met
intervals: 95% [−0.008, 0.008];		98.33% [−0.010, 0.010]							
PS2: swap, 15% of codes	500	0.870	0.876	8	5	13	0.006	0.581 (1.00)	not rejected; positive control not met
intervals: 95% [−0.008, 0.020];		98.33% [−0.012, 0.024]							

Columns as in Table S5 (b: items that GATE-M answers correctly and the ungated rule incorrectly; c: the reverse; $n_d = b + c$; net difference $= (b - c)/n$); Holm over two; 95% and 98.33% paired percentile bootstrap intervals (10 000 resamples). A pipeline check, not read beside family P; the control was not met and the plan's audit found no defect. Source: runs/P2-GATE-M4/analysis.json (ps.per_corruption.*, ps.positive_control); runs/P2-A23-AUDIT/a23_ps_audit.json.

E. Token Detection: Values of the Main Paper's Detection Figure

Table S11 gives the values plotted in Fig. 5 of the main paper.

TABLE S11
DETECTION OF CHANGED IMAGE TOKENS ON THE DETECTION ITEMS: PER-ITEM AUROC

Corruption	Items	q (AUROC)	TbV score, one pass (AUROC)	Best cheap rival, sign (AUROC)	q minus rival
Patch, 8×8 codes	200	0.961 [0.959, 0.963]	0.854 [0.847, 0.861]	plain B (+): 0.964 [0.962, 0.966]	−0.003 [−0.004, −0.003]
Swap, 15% of codes	200	0.955 [0.952, 0.957]	0.859 [0.854, 0.864]	plain B (+): 0.960 [0.958, 0.962]	−0.005 [−0.006, −0.005]
JPEG, quality 10	197	0.561 [0.547, 0.575]	0.581 [0.569, 0.593]	patch texture (−): 0.756 [0.743, 0.768]	−0.195 [−0.214, −0.175]
Noise, $\sigma = 0.1$	198	0.503 [0.493, 0.513]	0.607 [0.597, 0.618]	patch texture (−): 0.768 [0.760, 0.776]	−0.265 [−0.278, −0.253]

Mean per-item AUROC against changed-code labels, 95% percentile bootstrap intervals over items (10 000 resamples). q: the gate's score; TbV score: one-pass observed-position score of the non-finetuned model; rival: the best cheap score in either orientation, chosen on these data (same-item selection, so optimistic), with its sign. Items: detection items entering each row. Last column: paired difference q minus rival (Holm-adjusted $p = 0.0004$ for JPEG and noise). Source: runs/P2-GATE-M1/analysis.json, keys table.*, paired_q_vs_best_cheap.*.

F. Equal-Size Gates (Descriptive) and the Evidence Ratio (S-Z)

Table S12 reports GATE-M against the other gates of equal size, descriptively. On every realistic corruption GATE-M was not distinguishable from the geometry-matched and scattered random gates; under splice the top-k gate on plain B gave a higher accuracy than GATE-M (−0.008, 95% interval −0.016 to −0.002); the changed-code gate lowered accuracy relative to no gate under JPEG (−0.084, −0.122 to −0.046) and raised it under noise (0.070, 0.036 to 0.104).

TABLE S12
EQUAL-SIZE GATES ON THE 500 TEST ITEMS

Corruption	GATE-M minus: geometry-matched	scattered	plain-B top-k	s_3 top-k	changed-code	Changed-code minus none
JPEG	0.000	−0.002	0.000	0.000	**0.082**	**−0.084**
	[−0.002, 0.002]	[−0.007, 0.001]	[−0.006, 0.006]	[−0.008, 0.008]	[0.044, 0.120]	[−0.122, −0.046]
Noise	0.001	0.000	−0.002	0.000	**−0.070**	**0.070**
	[0.000, 0.002]	[0.000, 0.000]	[−0.006, 0.000]	[0.000, 0.000]	[−0.104, −0.036]	[0.036, 0.104]
Splice	−0.001	−0.004	**−0.008**	−0.006	−0.008	0.004
	[−0.009, 0.007]	[−0.011, 0.002]	[−0.016, −0.002]	[−0.016, 0.002]	[−0.022, 0.006]	[−0.010, 0.018]

Accuracy of GATE-M minus each comparison gate, and of the changed-code gate minus no gate, with paired percentile bootstrap 95% intervals over items (10 000 resamples). Geometry-matched and scattered random gates are averaged over three seeds per item. Descriptive (exploratory); no p-value. Bold: the 95% interval excludes zero. Source: runs/P2-GATE-M4/analysis.json, keys mechanism.{jpeg10,noise0.1,splice128}.y.*.

GATE-M increased the evidence ratio (mean over spread of the kept Δ_j) under all three corruptions (Wilcoxon signed-rank, Holm over three): by 0.0090 (0.0078 to 0.0102; $p_{\text{Holm}} = 1.9 \times 10^{-39}$) under JPEG, 0.0038 (0.0030 to 0.0047; 8.5×10^{-16}) under noise and 0.0078 (0.0069 to 0.0087; 2.9×10^{-43}) under splice. This is not a result about verification accuracy: the ratio rises when tokens with extreme evidence are removed for any reason. The geometry-matched random gate did not raise it (GATE-M minus that gate 0.0091, 0.0040, 0.0081), the plain-B top-k gate raised it almost as much as GATE-M (differences 0.0008, 0.0003, 0.0005), and the uncertainty top-k gate less (0.0051, 0.0033, 0.0045).

G. Pixel Denoising (S-DN)

The denoiser is OpenCV non-local means with one setting for all corruptions; on 20 images it raises the peak signal-to-noise ratio of the noisy image from 20.41 to 25.75 dB and lowers that of JPEG quality 10 from 26.82 to 25.13 dB. Under noise the denoised copy gave higher verification accuracy than GATE-M: 0.852 against 0.676 (GATE-M minus denoised −0.176, 95% interval −0.224 to −0.130; $p_{\text{Holm}} = 1.5 \times 10^{-12}$). Under JPEG no difference was detected (0.844 against 0.836; −0.008, −0.042 to 0.026; $p_{\text{Holm}} = 0.74$); at this setting the denoiser harms JPEG images, so this row compares GATE-M with a defence that lowers image quality.

H. Decomposition (E-dec) and Sign Agreement (E-pred), Descriptive

E-dec splits the accuracy lost to corruption into the residual loss under the changed-code (oracle) gate, the recovery by GATE-M, and the gap to the oracle gate (Table S13); its parts are relative to the changed-code label, which marks encoder instability and is not an evidence oracle. E-pred compares the committed sign of the independent-token signal-to-noise change with the observed sign of $b - c$ (Table S14): agreement in 2 of 2 evaluable primary cells (95% interval 0.158 to 1.000), the noise cell excluded for having no observed sign, and in 0 of 1 positive-control cells; the rival rule "always +" agreed in 0 of 2. No cell had an observed interval that excludes zero, so the plan's refutation rule was not met. On the evidence ratio, which GATE-M raised, the committed sign was negative; this disagreement is reported and not read either way. The correlated-token proposition of the main paper is not used to interpret E-pred.

TABLE S13
E-DEC ON THE ACCURACY SCALE, PRIMARY CELLS (DESCRIPTIVE; NO TEST)

Cell	n	T	U	R	V
JPEG (P1)	498	**0.044** [0.018, 0.070]	**0.124** [0.088, 0.161]	−0.002 [−0.006, 0.000]	**−0.078** [−0.116, −0.040]
Noise (P2)	500	**0.204** [0.166, 0.244]	**0.134** [0.098, 0.172]	0.000 [0.000, 0.000]	**0.070** [0.036, 0.106]
Splice (P3)	500	0.004 [−0.006, 0.014]	0.000 [−0.016, 0.016]	−0.004 [−0.012, 0.004]	0.008 [−0.006, 0.024]

T: clean minus corrupted accuracy without a gate; U: residual loss under the oracle (changed-code) gate; R: realised recovery by GATE-M; V: gap to the oracle gate; $T = U + R + V$. Paired percentile bootstrap 95% intervals over items, items resampled jointly (10 000 resamples). n: complete items (two JPEG items have no finite changed-code value). Bold: the 95% interval excludes zero. Source: runs/P2-GATE-M4/edec.json, keys cells.{P1,P2,P3}.accuracy_scale.tau.*.

TABLE S14
E-PRED, PRIMARY CELLS (DESCRIPTIVE; FAMILY E WITHDRAWN, NO TEST)

Cell	predicted sign s_c	Same-sign share	b	c	Net diff. [95%]	observed sign o_c	o_c^*	Agree
JPEG (P1)	–	1.000	0	1	−0.002 [−0.006, 0.000]	–	none	yes
Noise (P2)	–	1.000	0	0	0.000 [0.000, 0.000]	none	none	excluded ($b = c$)
Splice (P3)	–	1.000	1	3	−0.004 [−0.012, 0.004]	–	none	yes

s_c: committed sign of the plug-in signal-to-noise change (the independent-token condition of Lemma 1 of the main paper) from the fit chunk, with the share of fit resamples giving the same sign; o_c: sign of $b - c$ for GATE-M against none; o_c^*: the same sign only where the 95% paired bootstrap interval of the net difference excludes zero. Net difference with its paired percentile bootstrap 95% interval over items. Agreement $A/K_{\text{eval}} = 2/2$ (Clopper–Pearson 95% interval 0.158 to 1.000); rival rules: "always +" 0/2, "− for primary cells, + for positive-control cells" 2/2. No estimate is bold: no interval excludes zero. Positive-control cells are reported in the text. Source: runs/P2-GATE-M4/epred_score.json, keys cells.{P1,P2,P3} and P.

I. Coherence of the Denoiser Across Mask Patterns

On 100 clean calibration items (800 gaps) the median absolute order gap of the joint log-likelihood was 0.071 nats, above the numerical noise floor (95th percentile of repeat differences 0.000) and below the clean mean per-token evidence $\mu = 0.199$, so neither pre-stated reading applies; by pair type (descriptive) the median gap was 0.253 for horizontally adjacent, 0.246 for vertically adjacent, and 0.032 for distant tokens.

J. Exploratory Corruption Suite

The suite has eight cells of 100 test items (Table S15); the four types and two severities were chosen by kind from a package of fifteen types with no recorded rationale. Intervals only; no p-value enters a claim.

TABLE S15
EXPLORATORY CORRUPTION SUITE: GATE-M AGAINST NO GATE (100 ITEMS PER CELL; INTERVALS ONLY)

Cell (imagecorruptions, severity)	Acc. none	Acc. GATE-M	b	c	Net [95%]
Pixelate, 2	0.82	0.82	0	0	0.00 [0.00, 0.00]
Pixelate, 4	0.76	0.76	0	0	0.00 [0.00, 0.00]
Defocus blur, 2	0.82	0.82	0	0	0.00 [0.00, 0.00]
Defocus blur, 4	0.75	0.76	1	0	0.01 [0.00, 0.03]
Fog, 2	0.77	0.76	0	1	−0.01 [−0.03, 0.00]
Fog, 4	0.81	0.80	0	1	−0.01 [−0.03, 0.00]
Impulse noise, 2	0.60	0.61	1	0	0.01 [0.00, 0.03]
Impulse noise, 4	0.55	0.55	0	0	0.00 [0.00, 0.00]

K. Re-Infilling (S2): Detail

In the flagged-set arm the four draws share one position set, while in the random-set arm each draw has its own random set, so the best-of-four selection acts over position sets as well as noise in the random-set arm; the direction of the net bias is not established. Re-infilling clean positions with caption-masked prior samples destroys caption evidence, so the random-set arm may fall below no gate. Against no gate, the flagged-set arm gave 0.003 (−0.007 to 0.013) under patch and −0.003 (−0.017 to 0.007) under splice (descriptive). The share of re-infilled positions whose code equals the original clean code was below 0.04 in every arm and cell (descriptive; not restoration). The sensitivity analysis on the first 150 items rejected no hypothesis in either cell.

L. Family B: Fit Pairs and Minimum Detectable Gain

On the 239 fit pairs the better direction was the text direction (accuracy 0.887, below the untestable ceiling of 0.95). The minimum detectable net gain at the test n of 298 pairs is 0.054 at an assumed discordance of 0.10.

M. Proof of Proposition 2 of the Main Paper (Withholding With Exchangeably Correlated Tokens)

This proof is theory under stated assumptions, not validated on UniDisc, and is not used in Study 2; the notation is that of Section IV-A of the main paper, and equation and figure numbers refer to the main paper. Write $m_0 = (1-\rho)m$ and $m_1 = \rho m$ for the numbers of clean and corrupted tokens, $k_0 = 1-\phi_0$ and $k_1 = 1-\phi_1$ for the keep probabilities, and $z_j \in \{0,1\}$ for the label of token j, so that $\mathbb{E}\Delta_j$ is μ or μ_c and $\operatorname{Var}\Delta_j$ is σ^2 or σ_c^2. The exchangeable block covariance of assumption (a) is positive semi-definite if and only if $\gamma_0 \geq -1/(m_0-1)$, $\gamma_1 \geq -1/(m_1-1)$, $\gamma_0, \gamma_1 \leq 1$ and $\gamma_{01}^2\, m_0 m_1 \leq \{1+(m_0-1)\gamma_0\}\{1+(m_1-1)\gamma_1\}$.

Variance: By assumption (b), I_j is independent of Δ_j, so $\operatorname{Var}(I_j\Delta_j) = k(\sigma_z^2+\mu_z^2) - k^2\mu_z^2 = k\sigma_z^2 + k(1-k)\mu_z^2$ with $k = k_{z_j}$, which are the terms of (9). For $j \neq k$, again by (b), $\operatorname{Cov}(I_j\Delta_j, I_k\Delta_k) = \mathbb{E}I_j\,\mathbb{E}I_k\,\mathbb{E}[\Delta_j\Delta_k] - \mathbb{E}I_j\,\mathbb{E}I_k\,\mathbb{E}\Delta_j\,\mathbb{E}\Delta_k = k_{z_j}k_{z_k}\operatorname{Cov}(\Delta_j,\Delta_k)$. There are $m_0(m_0-1)$ ordered clean pairs with covariance $\gamma_0\sigma^2$, $m_1(m_1-1)$ ordered corrupted pairs with covariance $\gamma_1\sigma_c^2$ and $2m_0m_1$ ordered mixed pairs with covariance $\gamma_{01}\sigma\sigma_c$, which gives (11).

Condition: Let V_g be (11) at (ϕ_0,ϕ_1) and V_0 its value at $\phi_0 = \phi_1 = 0$. With $\mu_c = 0$, $\mathbb{E}S_{\mathrm{G}} = m_0k_0\mu$ with the gate and $m_0\mu$ without it, so, both being positive, the gate raises the ratio if and only if $k_0^2V_0 > V_g$. Collect $k_0^2V_0 - V_g$ by type of term. The clean terms are $k_0^2m_0\sigma^2\{1+(m_0-1)\gamma_0\} - m_0k_0(\sigma^2+\phi_0\mu^2) - m_0(m_0-1)k_0^2\gamma_0\sigma^2 = -m_0k_0\phi_0(\sigma^2+\mu^2)$, in which the γ_0 terms cancel. The corrupted terms are $m_1\sigma_c^2\{(k_0^2-k_1)+(m_1-1)\gamma_1(k_0^2-k_1^2)\}$, with $k_0^2-k_1 = \phi_1-2\phi_0+\phi_0^2$ and $k_0^2-k_1^2 = (\phi_1-\phi_0)(2-\phi_0-\phi_1)$. The mixed terms are $2m_0m_1\gamma_{01}\sigma\sigma_c\,k_0(k_0-k_1)$, with $k_0-k_1 = \phi_1-\phi_0$. Dividing the sum by m gives (12).

Consequences: (i) is read off (12). (ii) At $\phi_1 = \phi_0 \in (0,1)$ every correlation term vanishes and the left side is $\rho\sigma_c^2(\phi_0^2-\phi_0) < 0$, below the positive right side. For $\phi_1 \leq \phi_0$, let $G = (\rho m-1)\gamma_1$, so that $G \geq -1$ by positive semi-definiteness. The bracket $(\phi_1-2\phi_0+\phi_0^2) + G(\phi_1-\phi_0)(2-\phi_0-\phi_1)$ is linear in G; it equals $\phi_1-2\phi_0+\phi_0^2 \leq -\phi_0(1-\phi_0)$ at $G=0$, equals $-\phi_1(1-\phi_1) \leq 0$ at $G=-1$, and does not increase for $G \geq 0$; hence it is at most 0. If $\gamma_{01} \geq 0$ the mixed term is at most 0 as well, while the right side is positive for $\phi_0 \in (0,1)$. (iii) The derivatives of the left side with respect to γ_1 and γ_{01} are $\rho\sigma_c^2(\rho m-1)(\phi_1-\phi_0)(2-\phi_0-\phi_1)$ and $2\rho(1-\rho)m\sigma\sigma_c(1-\phi_0)(\phi_1-\phi_0)$, both positive when $\phi_1 > \phi_0$ and $\rho m > 1$, and the right side does not depend on γ_1 or γ_{01}. For fixed $\phi_1 > \phi_0$ the γ_1 term grows without bound with $(\rho m-1)\gamma_1$, which gives the limit. With $\gamma_{01} = 0$, $x = \phi_1-\phi_0$ and $\chi = \rho\sigma_c^2/\{(1-\rho)(\sigma^2+\mu^2)\}$, (12) reads $Gx^2 - x\{1+2G(1-\phi_0)\} + \phi_0(1-\phi_0)(1+1/\chi) < 0$, so for $G > 0$ the gate raises the ratio if and only if ϕ_1 exceeds $b(\phi_0; G, \chi) = \phi_0 + \big[1+2G(1-\phi_0) - \sqrt{\{1+2G(1-\phi_0)\}^2 - 4G\phi_0(1-\phi_0)(1+1/\chi)}\big]/(2G)$ (and $\phi_1 \leq 1$), the curve of Fig. 3; as $G \to 0$ it tends to the boundary $2\phi_0 - \phi_0^2 + \phi_0(1-\phi_0)/\chi$ of (10). (iv) For $\gamma_{01} < 0$ and $\phi_1 < \phi_0$ the mixed term is positive and grows with m, so it can exceed the remaining terms. An instance with a positive definite covariance is $m_0 = 900$, $m_1 = 100$, $\gamma_0 = \gamma_1 = 0.05$, $\gamma_{01} = -0.045$, $\mu = 0.1$, $\mu_c = 0$, $\sigma^2 = \sigma_c^2 = 1$: the smallest eigenvalue of the correlation matrix is 0.9500, and the ratio rises from 0.4892 without a gate to 0.5080 with the gate $(\phi_0,\phi_1) = (0.3, 0)$. The instance illustrates the algebra and is not an estimate from the model. (v) m enters (12) through $\rho m - 1$ and m whenever $\gamma_1 \neq 0$ or $\gamma_{01} \neq 0$. □

N. Cost of Study 1

TABLE S16
Cost per Item of the Main-Test Stage of Study 1

Method (stage)	Items	Network rows per item	Compute time per item, s	Peak GPU memory, GB
NONE and GATE-M (main-test pass; all Study 1 arms share it)	500	84 (84–86)	11.2 (11.0–11.5)	8.2

One laptop GPU (12 GB). Network rows: median (range); time: median (quartiles) of the busy time per item, excluding duty-cycle pauses; memory: maximum over items. All arms share the pass, so per-arm cost is not measured separately. Source: runs/P2-GATE-M4/items.jsonl, fields n_rows, seconds, peak_gb.

O. Exploratory (Tier 3): Hard Gate Against Detector-Free Combiners, Discounting, and Reliability Weighting of the Same Evidence

This comparison was pre-specified in Addendum A4 and is exploratory (Tier 3): intervals only, no p-value, 72 intervals, and no multiplicity adjustment; it was run once after the primary analysis was committed, on the same Δ_j, items, and decision rule, and it cannot explain, qualify, or rescue the primary result. Each comparator is contrasted with the hard gate (GM, which is GATE-M). The reading of a contrast comes from the exact Clopper–Pearson 95% interval for the share of discordant items won by the comparator: *above* GM if its lower end exceeds one half, *below* if its upper end is under one half, *identical* if $n_d = 0$, and *not distinguishable* otherwise. Beside it are the net difference with a

descriptive bootstrap interval and the bound used for added value: the exact 95% upper bound B on the discordance rate for $n_d < 10$, and the minimum detectable difference (MDD) for $n_d \geq 10$.

No detector-free combiner was detectably above the hard gate on a realistic corruption; under noise the median and the 20% trimmed mean scored 0.678 and 0.682 against 0.676 for the hard gate, with intervals of the difference that contain zero (Table S17). The hard gate gave higher verification accuracy than the median under JPEG (−0.032, −0.052 to −0.014; exact interval for the discordant split 0.047 to 0.374) and splice (−0.032, −0.050 to −0.016; 0.001 to 0.273), and than the 20% trimmed mean under JPEG (−0.020; 0.018 to 0.428) and splice (−0.026; 0.000 to 0.247). The data do not show a difference between the hard gate and the trimmed mean at the gate's flag count under JPEG ($n_d = 5$; $B = 0.023$), noise ($n_d = 2$; $B = 0.014$), or splice ($n_d = 3$; $B = 0.017$), nor, under noise, between the hard gate and the median ($n_d = 19$; MDD 0.025) or the 20% trimmed mean ($n_d = 9$; $B = 0.034$); because each bound is at most 0.05, each counts, by the pre-stated rule, as a finding against the added value of the detector over a detector-free combiner on that cell. By limit (c) of Table S17, the contrast with the trimmed mean at the gate's flag count largely repeats the primary contrast when the gate's score is near chance. Discarding and discounting the flagged tokens were not distinguishable under JPEG or splice and made the same decision on all 500 items under noise; discarding and reliability weighting of every token were not distinguishable on any realistic corruption ($n_d = 15$, 18, and 12; MDD 0.022, 0.024, and 0.020). On the positive-control cells (Table S18; pipeline check only), reliability weighting of every token gave higher accuracy than the hard gate under swap (0.016, 0.004 to 0.030; exact interval 0.516 to 0.979); this concerns discarding against weighting on a sanity corruption and is not a finding about the realistic corruptions.

TABLE S17

Exploratory (Tier 3): The Hard Gate GM (GATE-M) Against Detector-Free Combiners, Discounting, and Reliability Weighting of the Same Evidence. Limits: (a) no sign-aware detector-free estimator is computed, because the direction in which a corruption moves Δ_j is defined by the true caption and such a combiner would use the label; (b) SOFT-G's weights lie token by token between GM's and no gate's, so it is not reliability weighting of every token; (c) when the gate's score is near chance the flagged count is close to the false-alarm count, so TM-G approximates no gate trimmed at about that count and TM-G minus GM largely repeats the primary contrast; (d) at this n many contrasts are uninformative, as the bound shows.

Cell	Comparator	Acc.	b/c	n_d	Δ [95%, descriptive]	Exact 95%	Bound	Reading
JPEG, quality 10	(GM 0.836; no gate 0.838; mean flagged tokens 61.8)							
	MED	0.804	4/20	24	**−0.032** [−0.052, −0.014]	0.047–0.374	MDD 0.027	below GM
	TM-G	0.834	2/3	5	−0.002 [−0.010, 0.006]	0.053–0.853	B 0.023	n.d.†
	TM20	0.816	2/12	14	**−0.020** [−0.034, −0.006]	0.018–0.428	MDD 0.021	below GM
	SOFT-G	0.838	1/0	1	0.002 [0.000, 0.006]	0.025–1.000	B 0.011	n.d.
	SOFT-A	0.838	8/7	15	0.002 [−0.014, 0.018]	0.266–0.787	MDD 0.022	n.d.
Noise, $\sigma = 0.10$	(GM 0.676; no gate 0.676; mean flagged tokens 33.9)							
	MED	0.678	10/9	19	0.002 [−0.016, 0.020]	0.289–0.756	MDD 0.025	n.d.†
	TM-G	0.676	1/1	2	0.000 [−0.006, 0.006]	0.013–0.987	B 0.014	n.d.†
	TM20	0.682	6/3	9	0.006 [−0.006, 0.018]	0.299–0.925	B 0.034	n.d.†
	SOFT-G	0.676	0/0	0	0.000	–	B 0.007	identical
	SOFT-A	0.676	9/9	18	0.000 [−0.016, 0.016]	0.260–0.740	MDD 0.024	n.d.
Splice, 128 px	(GM 0.872; no gate 0.876; mean flagged tokens 46.0)							
	MED	0.840	1/17	18	**−0.032** [−0.050, −0.016]	0.001–0.273	MDD 0.024	below GM
	TM-G	0.866	0/3	3	−0.006 [−0.014, 0.000]	0.000–0.708	B 0.017	n.d.†
	TM20	0.846	0/13	13	**−0.026** [−0.040, −0.014]	0.000–0.247	MDD 0.020	below GM
	SOFT-G	0.876	2/0	2	0.004 [0.000, 0.010]	0.158–1.000	B 0.014	n.d.
	SOFT-A	0.880	8/4	12	0.008 [−0.006, 0.022]	0.349–0.901	MDD 0.020	n.d.
Clean pairs, cost of each combiner	(GM 0.882; no gate 0.880; mean flagged tokens 46.7)							
	MED	0.852	1/16	17	**−0.030** [−0.046, −0.014]	0.001–0.287	MDD 0.023	below GM
	TM-G	0.874	1/5	6	−0.008 [−0.018, 0.000]	0.004–0.641	B 0.026	n.d.
	TM20	0.854	0/14	14	**−0.028** [−0.044, −0.014]	0.000–0.232	MDD 0.021	below GM
	SOFT-G	0.880	0/1	1	−0.002 [−0.006, 0.000]	0.000–0.975	B 0.011	n.d.
	SOFT-A	0.884	6/5	11	0.002 [−0.012, 0.014]	0.234–0.833	MDD 0.019	n.d.

Detector-free combiners: MED, median; TM-G, trimmed mean removing $\lfloor g_i/2 \rfloor$ tokens from each tail, g_i the gate's flag count; TM20, 20% trimmed mean. SOFT-G: discounting of the gate-flagged tokens; SOFT-A: reliability weighting of every token. All on the same Δ_j and items ($n = 500$ per cell; primary cells and the clean cell). Acc.: accuracy of the comparator. b/c: items right only under the comparator / only under GM. Δ: comparator minus GM with its percentile bootstrap 95% interval (descriptive; 10 000 resamples, one index matrix per cell). Exact: Clopper–Pearson 95% interval for b/n_d, which gives the reading. Bound: B (exact 95% upper bound on the discordance rate) for $n_d < 10$, MDD for $n_d \geq 10$. Reading: above or below GM, not distinguishable (n.d.), or identical; †: counts as a finding against the added value of the detector (n.d. or identical for a detector-free combiner with bound at most 0.05; primary cells only). Bold: the exact interval excludes one half. 72 intervals in this comparison; no p-value and no multiplicity adjustment. Source: runs/P2-GATE-M4/tier3_fusion.json, keys cells.*.arms, cells.*.contrasts.*_minus_GM, clean.arms, clean.contrasts.

Theory reference for Study 1's soft comparators; not used in Study 2: The following statements are theory under stated assumptions, not validated on UniDisc. They describe what the soft comparators above approximate under a model; they imply no ranking of the comparators, the optimal weight below is not a comparator and was not computed on any data, and no sentence of the paper combines them with the correlated-token proposition of the main paper

TABLE S18
EXPLORATORY (TIER 3), POSITIVE-CONTROL CELLS ONLY

Cell	Comparator	Acc.	b/c	n_d	Δ [95%, descriptive]	Exact 95%	Bound	Reading
Patch	(GM 0.870; no gate 0.870; mean flagged tokens 96.5)							
	MED	0.846	2/14	16	$-\mathbf{0.024}$ $[-0.040, -0.010]$	0.016–0.383	MDD 0.022	below GM
	TM-G	0.858	2/8	10	-0.012 $[-0.024, 0.000]$	0.025–0.556	MDD 0.018	n.d.
	TM20	0.846	2/14	16	$-\mathbf{0.024}$ $[-0.040, -0.010]$	0.016–0.383	MDD 0.022	below GM
	SOFT-G	0.870	0/0	0	0.000	–	B 0.007	identical
	SOFT-A	0.880	8/3	11	0.010 $[-0.002, 0.024]$	0.390–0.940	MDD 0.018	n.d.
Swap	(GM 0.876; no gate 0.870; mean flagged tokens 209.7)							
	MED	0.838	1/20	21	$-\mathbf{0.038}$ $[-0.056, -0.020]$	0.001–0.238	MDD 0.025	below GM
	TM-G	0.852	1/13	14	$-\mathbf{0.024}$ $[-0.040, -0.010]$	0.002–0.339	MDD 0.021	below GM
	TM20	0.846	0/15	15	$-\mathbf{0.030}$ $[-0.046, -0.016]$	0.000–0.218	MDD 0.021	below GM
	SOFT-G	0.878	2/1	3	0.002 $[-0.004, 0.010]$	0.094–0.992	B 0.017	n.d.
	SOFT-A	0.892	10/2	12	$\mathbf{0.016}$ $[0.004, 0.030]$	0.516–0.979	MDD 0.019	above GM

Pipeline check; never pooled with or shown beside the primary cells. Columns, comparators, bound, reading, and bold rule as in Table S17; $n = 500$ per cell; "against added value" is not read on these cells. Source: runs/P2-GATE-M4/tier3_fusion.json, keys ps_cells.{patch,swap}.arms and ps_cells.{patch,swap}.contrasts.*_minus_GM.

(whose assumptions differ, see Remark (iii) below). Write the soft comparators' weights as $\omega_j \geq 0$ and $S_\omega = \sum_j \omega_j \Delta_j$; the reported normalised mean has the sign of S_ω.

Lemma S1 (weighted statistic)**.** *Assume the moments and correlations of the main paper's correlated-token proposition, and that within a class the ω_j are i.i.d. with mean $\bar{\omega}_z$ and second moment $\overline{\omega^2}_z$, independent across classes and of all Δ. With m_0 clean and m_1 corrupted tokens,*

$$\mathbb{E} S_\omega = m_0 \bar{\omega}_0 \mu + m_1 \bar{\omega}_1 \mu_c, \tag{S1}$$

$$\begin{aligned} \operatorname{Var} S_\omega = {} & m_0\{\overline{\omega^2}_0(\sigma^2 + \mu^2) - \bar{\omega}_0^2 \mu^2\} + m_1\{\overline{\omega^2}_1(\sigma_c^2 + \mu_c^2) - \bar{\omega}_1^2 \mu_c^2\} \\ & + m_0(m_0 - 1)\bar{\omega}_0^2 \gamma_0 \sigma^2 + m_1(m_1 - 1)\bar{\omega}_1^2 \gamma_1 \sigma_c^2 + 2 m_0 m_1 \bar{\omega}_0 \bar{\omega}_1 \gamma_{01} \sigma \sigma_c. \end{aligned} \tag{S2}$$

The hard gate is $\omega = I$ and returns the variance of the main paper.

Proposition S1 (signal-to-noise-optimal weights as a function of the score)**.** *Condition on the scores $u_1, \ldots, u_m$ and let $\omega_j = \omega(u_j)$. Assume (W1) Δ_j is independent of u_j given its label; (W2) the labels are independent across tokens given the scores; (W3) $\sigma_c = \sigma$ and one exchangeable correlation γ for all pairs. With $\pi_1(u)$ the probability that a token with score u is corrupted, $\mu(u) = (1 - \pi_1(u))\mu + \pi_1(u)\mu_c$, $\sigma^2(u) = \sigma^2 + \pi_1(1 - \pi_1)(\mu - \mu_c)^2$ and $d_j = \sigma^2(u_j) - \gamma\sigma^2$: (a) for $\gamma = 0$ and $\omega \geq 0$ the optimum is $(\mu(u))_+/\sigma^2(u)$, and a 0/1 weight is optimal if and only if this function takes at most one non-zero value on the support of u (up to scale); with $\mu_c = 0$ and $\mu^2 \ll \sigma^2$ it is approximately proportional to $1 - \pi_1(u)$; (b) for $\gamma > 0$ and $\omega \geq 0$ the optimum is $\omega^*(u_j) \propto (\mu(u_j) - \xi)_+/d_j$ with $\xi = \gamma\sigma^2 \sum_k (\mu(u_k) - \xi)_+/d_k$, which has a unique solution $\xi \geq 0$; if some $\mu(u_k) > 0$, then $\xi = 0$ if and only if $\gamma = 0$, and if every $\mu(u_k) \leq 0$, then $\xi = 0$ for every γ; correlation together with non-negativity induces a cut; (c) without $\omega \geq 0$ the optimum is $\Sigma^{-1}\mu(u)$, which has no cut and gives negative weights to tokens with low $\mu(u)$.*

Proof of (b). Maximising the ratio over the cone $\omega \geq 0$ is, up to scale, the convex problem $\min_{\omega \geq 0} \frac{1}{2}\omega^\top \Sigma \omega - \omega^\top \mu(u)$ with $\Sigma = D + \gamma\sigma^2 \mathbf{1}\mathbf{1}^\top$; its Karush–Kuhn–Tucker conditions give $\omega_j = (\mu(u_j) - \xi)/d_j$ on the support with $\xi = \gamma\sigma^2 \sum_k \omega_k$, and the right side of the fixed-point equation is continuous and non-increasing in ξ while the left side increases. (a) follows from the Cauchy–Schwarz inequality. □

Corollary S1 (cost on a clean pair)**.** *On a clean pair ($\rho = 0$) any non-constant weight lowers the ratio. Given the weights, with $\nu = m \sum_j \omega_j^2/(\sum_j \omega_j)^2$ and clean correlation γ_0, the squared ratio relative to no weighting is $\{(1 - \gamma_0) + \gamma_0 m\}/\{(1 - \gamma_0)\nu + \gamma_0 m\}$.*

Remarks. (i) For weights spread uniformly over a conformal p-value grid the limit of this ratio at $\gamma_0 = 0$ is 3/4 (a ratio of squared signal-to-noise ratios), but within an item the scores are clustered, so 3/4 is not the expected clean-cell cost of the per-token weighting comparator and is not compared with the clean cell above. (ii) The hard gate is a 0/1 weight; under (a) it is optimal only in the case stated there. (iii) (W3) forces $\gamma_0 = \gamma_1 = \gamma_{01}$ and $\sigma_c = \sigma$, which is not the case in which the main paper's frontier moves, so the two statements are not jointly applicable.

S-III. STUDY 1: FAMILY H (ADDED AFTER THE PRIMARY RESULT)

In a separately labelled family added after the primary result was known (family H: SugarCrepe hard negatives on 50 pilot COCO images not used elsewhere; Holm over the same three corruptions), the pre-written pilot rule found no corruption with decision headroom on 50 pilot images, so the family's test (H4) was not run, and nothing

is stated about the gate on hard negatives. Table S19 gives the recorded decision; the item values of the pilot are in runs/P2-GATE-H3/family.json and are not interpreted here.

TABLE S19
Family H (Added After the Primary Result): Recorded Decision

Item	Recorded decision
Preregistered decision rule	H-NOGO
Test performed	no
Downstream claim	withdrawn

In a separately labelled family added after the primary result was known (family H: SugarCrepe hard negatives on 50 pilot COCO images not used elsewhere; Holm over the same three corruptions). Decision fields only. Source: runs/P2-GATE-H3/family.json, keys n_rule.h_rule, go, downstream_claim_withdrawn.

Disclosure: In a separately labelled family added after the primary result was known (family H), the following applies. Family H was added on 29 October 2026, after the following had been computed and committed: the primary family (label negative) with its secondary families and mechanism arms (S-M, S-Z, S-DN, S-D2, S-C), the calibration check, and the clean-accuracy look; the positive-control family PS and the pipeline audit it triggered (no defect found); the decomposition and sign-agreement analyses (E-dec, E-pred); and the exploratory comparators (Tier 3). Of these, the analyst who designed family H read the primary family, S-M, the completeness and clean-accuracy records, parts of the mechanism and calibration sections, the positive-control audit, and a one-line summary of the Tier-3 result; the E-dec and E-pred outputs had been committed but were not opened by that analyst. Family H tests the same contrast on a different population of image-caption pairs (SugarCrepe hard negatives on COCO images used nowhere else in this study). It was specified in Addendum A5 of the statistical analysis plan (draft `86064e57`; final `b32e72a1`; revision r1 `2f62f1df`) before any hard-negative item was scored, approved by the first author (decision D0080), signed with conditions by the adversarial review, and implemented in audited code (commit `5e6380b1`). It reuses the primary family's corruptions, severities, threshold, and arms. Its sample-size rule uses the gate's own unsigned discordance on 50 separate pilot images. Family H is the last confirmatory family added after the primary result; it has its own Holm correction, cannot change the primary result, and is reported whatever its outcome, including "not run".

Caveats: In a separately labelled family added after the primary result was known (family H), the following caveats apply to its pool. SugarCrepe positives are original COCO val2017 captions of their images, and whether UniDisc saw them in training is unmeasured; SugarCrepe filtered its negatives against text-only models, not against UniDisc; the same image and positive recur across subsets, so one row per image is used; and the H images are a selected subset of val2017.

S-IV. Study 2: Exploratory and Descriptive Material

A. *Every Quantity Reported With Each Preregistered Test*

Table S20 gives, for every preregistered test of Study 2, every quantity that the analysis plan reports with each test. These are preregistered outputs, not exploratory material.

TABLE S20
Every Preregistered Test of Study 2 With All Reported Quantities

Test	n	b	c	n_d	Acc. none	Acc. LRF	Net	Decision
L1: noise, $\sigma = 0.1$	450	58	27	85	0.733	0.802	**0.069**	**confirmed (one-sided)**
90% [0.036, 0.102]; 95% [0.029, 0.109]; Clopper–Pearson 95% of b/n_d [0.572, 0.779]; exact p: one-sided 0.0005, two-sided 0.001; mid-p: one-sided 0.0004, two-sided 0.0007; Holm p not applicable								
L3: JPEG, quality 10	450	27	20	47	0.849	0.864	0.016	not rejected
90% [−0.009, 0.040]; 95% [−0.013, 0.044]; Clopper–Pearson 95% of b/n_d [0.422, 0.717]; exact p: one-sided 0.191, two-sided 0.382; mid-p: one-sided 0.156, two-sided 0.312; Holm p 0.382								
L4: splice, 128 px	450	24	10	34	0.896	0.927	**0.031**	**rejected (Holm)**
90% [0.009, 0.053]; 95% [0.007, 0.058]; Clopper–Pearson 95% of b/n_d [0.525, 0.849]; exact p: one-sided 0.012, two-sided 0.024; mid-p: one-sided 0.008, two-sided 0.017; Holm p 0.049								
L-NI: clean images	450	27	13	40	0.887	0.918	0.031	non-inferior: lower 90% bound 0.009 above −0.03
90% [0.009, 0.053]; 95% [0.004, 0.058]; Clopper–Pearson 95% of b/n_d [0.509, 0.814]; exact p: one-sided 0.019‡, two-sided 0.038‡; mid-p: one-sided 0.014‡, two-sided 0.028‡; Holm p not applicable								
L-LEN: noise, length-matched	449	67	33	100	0.708	0.784	**0.076**	**rejected (one-sided)**
90% [0.040, 0.111]; 95% [0.033, 0.120]; Clopper–Pearson 95% of b/n_d [0.569, 0.761]; exact p: one-sided 0.0004, two-sided 0.0009; mid-p: one-sided 0.0003, two-sided 0.0006; Holm p not applicable								
LG: shot noise	300	43	15	58	0.673	0.767	**0.093**	**rejected (Holm, one-sided)**
90% [0.053, 0.137]; 95% [0.047, 0.143]; Clopper–Pearson 95% of b/n_d [0.610, 0.847]; exact p: one-sided 0.0002, two-sided 0.0003; mid-p: one-sided 0.0001, two-sided 0.0002; Holm p 0.0005								
LG: impulse noise	300	35	20	55	0.700	0.750	**0.050**	**rejected (Holm, one-sided)**
90% [0.010, 0.090]; 95% [0.000, 0.100]; Clopper–Pearson 95% of b/n_d [0.496, 0.762]; exact p: one-sided 0.029, two-sided 0.058; mid-p: one-sided 0.022, two-sided 0.044; Holm p 0.029								
LG: speckle noise	300	32	14	46	0.697	0.757	**0.060**	**rejected (Holm, one-sided)**
90% [0.023, 0.097]; 95% [0.017, 0.103]; Clopper–Pearson 95% of b/n_d [0.542, 0.823]; exact p: one-sided 0.006, two-sided 0.011; mid-p: one-sided 0.004, two-sided 0.008; Holm p 0.011								

b: items LRF answers correctly and the unweighted rule incorrectly; c: the reverse; $n_d = b + c$; net $= (b - c)/n$. Intervals: paired percentile bootstrap over items (10 000 resamples). ‡ L-NI: p-values against zero net change for completeness only; the preregistered test of L-NI is the lower 90% bound against the margin −0.03, and superiority on clean images was not preregistered. Impulse noise: the two-sided p and the lower 95% bound 0.0 are stated in the main paper; no two-sided claim is made. Bold: null rejected at the preregistered level.

B. Item-Level Margins on the Fresh Fold (Descriptive)

Figure S1 shows the discordant items of L1 and of the clean cell by margin. It is descriptive, and superiority on clean images was not a preregistered hypothesis.

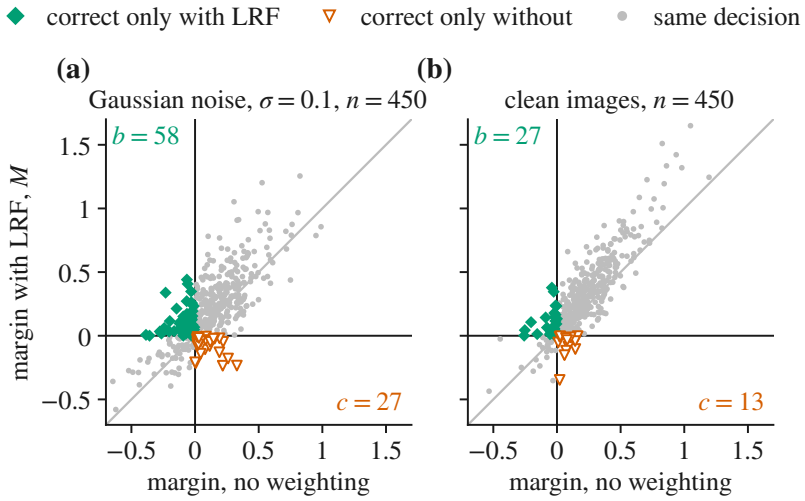


Fig. S1. Study 2, item-level margins on the fresh fold (descriptive; no test). Each point is one of the $n = 450$ items: horizontal, the margin without weighting, S/m; vertical, the LRF margin M (14) of the main paper. A decision is correct when its margin is positive (ties wrong), so the upper-left quadrant holds the b items correct only with LRF (filled diamonds) and the lower-right the c items correct only without weighting (open triangles); grey: same decision; line: equal margins. (a) Gaussian noise, standard deviation 0.1 (L1). (b) Clean images (L-NI), same items; descriptive only, and superiority on clean images was not a preregistered hypothesis. Counts b, c as in Table II of the main paper.

C. Exploratory and Descriptive Material

Everything in the rest of this section is exploratory or descriptive. The design-item numbers below are post hoc: they set the preregistered sample-size rule only and are not a performance estimate of the frozen object, which has no out-of-sample estimate on the design items.

Sample-size rule: The rule (Addendum A6, A6.3) sets Δ_{plan} to half of a cross-fitted design-item difference and takes the smallest n on a step-10 grid with exact one-sided McNemar power at least 0.8 at level 0.05, bounded to $[400, 533]$; its inputs and output are in Table S21.

TABLE S21
INPUTS AND OUTPUT OF THE SAMPLE-SIZE RULE OF STUDY 2 (POST HOC DESIGN-ITEM CROSS-FIT; SETS THE RULE ONLY)

Quantity	Value
Cross-fitted design items, noise: b, c, n (5 outer, 3 inner folds)	86, 28, 500
Δ_{cf}; discordance π_{cf}; $\Delta_{\text{plan}} = \Delta_{\text{cf}}/2$	0.116; 0.228; 0.058
n_{req}; n_L	450; 450
Power at n_L for Δ_{plan}, one-sided 0.05; 0.025	0.803; 0.704

Fresh-fold construction: The fold builder read every image set of the project and the seven SugarCrepe files (1 560 images) and excluded their union; images without instance annotations were not eligible. The primary wrong caption follows the derangement rule of the project's fold code; the length-matched wrong caption has the true caption's token count in 531 of 533 items and differs by one token in the other two.

Length strata of L1 (descriptive): By length stratum the L1 net change was 0.038 when the true caption was shorter than the wrong one (184 items), 0.140 when equal (43), and 0.081 when longer (223); longer minus shorter was 0.043 (95% interval −0.038 to 0.123).

Drop of the unweighted rule under the other noise types (descriptive): On the first 300 fold items the unweighted rule lost 0.217 (95% interval 0.163 to 0.270) under shot noise, 0.190 (0.143 to 0.240) under impulse noise, and 0.193 (0.143 to 0.247) under speckle noise, relative to clean images.

Fresh-fold GATE-M arm (descriptive, separate): The fresh-fold stage also computed Study 1's GATE-M arm, at the fixed caption slot with the threshold calibrated at the true-caption layout, so it carries no guarantee. Table S22 reports it on its own, with no LRF number; it is descriptive and enters no claim of either study.

Cost of Study 2: Table S23 gives the cost of the fresh-fold stages; the unweighted rule is the only reference.

TABLE S22
Fresh-Fold GATE-M Arm Against No Gate (Descriptive, No Guarantee, No Test; 450 Items)

Condition	Acc. none	Acc. GATE-M	b	c
Clean	0.887	0.889	2	1
JPEG, quality 10	0.849	0.847	0	1
Gaussian noise, $\sigma = 0.1$	0.733	0.727	0	3
Splice, 128 px	0.896	0.893	0	1

TABLE S23
Cost per Item of the Fresh-Fold Stages of Study 2

Method (stage)	Items	Network rows per item	Compute time per item, s	Peak GPU memory, GB
NONE, all four conditions (fresh-fold pass; the same pass stores the inputs of LRF)	450	67	8.0 (7.9–8.4)	8.2
NONE, three other noise types (LG pass)	300	42	5.0 (5.0–5.1)	8.2
LRF: weights and decision from stored arrays (CPU)	–	not measured	not measured	not measured

As for Table S16. LRF reads stored arrays on the CPU and adds no network row; no timing was recorded. Source: runs/P2-LRF-L1/items.jsonl, runs/P2-LRF-LG/items.jsonl.

D. Weight-Model Search, Caption-Slot Checks, and the Full Feature Table (Main Paper, Section V-B)

Nine configurations of $(Q, \lambda,$ learning rate) were drawn at random and pruned by synchronous successive halving (factor 3; 4, 13, and 40 epochs) on three inner folds of the 500 design items. On five items under four conditions the caption-masked scores were identical whether the slot was built from the true caption, the wrong caption, or padding, and the scorer reproduced the stored values at the true length (maximum absolute difference 0 in both checks). Any non-finite feature, weight, or margin stops the scorer and is treated as a pipeline defect, never as a wrong decision. Table S24 gives every feature by name and formula; Fig. 6(a) of the main paper groups them.

TABLE S24
Input Features of the Frozen Weight Model, by Name, Group, and Formula

Group	Count	Features	Definition
token scores	7	`B`, `D`, `q`, `s3`, `tbv`, `ltex`, `nbr`	B_j: log-probability of the observed code at token j with the tokens of j's group masked, the other image tokens visible and the caption masked; D_j: the same with every token masked; $r_j = B_j - D_j$; `s3` $= 1 - \max_y p(y)$ of the pass that gives B_j; `tbv`: log-probability of the observed code from one raw-logit pass with all image tokens visible and the caption masked; `ltex` $= \log(1 + v_j)$, v_j the pixel variance of the 16×16 patch of token j; `nbr`: mean L_2 distance of the codebook vector of token j to those of its eight neighbours
position	3	`row`, `col`, `rad`	`row`, `col`: grid row and column of the token divided by 31; `rad` $= \sqrt{(\text{row} - \frac{1}{2})^2 + (\text{col} - \frac{1}{2})^2}$
neighbourhood	8	`nb3mean_q`, `nb3mean_B`, `nb3mean_s3`, `nb3mean_tbv`, `nb3mean_ltex`, `nb3mean_nbr`, `nb3min_q`, `nb5mean_q`	`nb3mean_`k: mean of score k over the eight neighbours (edge-aware, token excluded); `nb3min_q`: their minimum of r; `nb5mean_q`: mean of r over the 5×5 neighbourhood without the token
within-image rank	4	`rank_q`, `rank_B`, `rank_s3`, `rank_tbv`	percentile rank of the score among the 1 024 tokens of the image, in $[0, 1]$
image mean	6	`itemmean_q`, `itemmean_B`, `itemmean_s3`, `itemmean_tbv`, `itemmean_ltex`, `itemmean_nbr`	mean of the score over the 1 024 tokens of the image
image standard deviation	6	`itemsd_q`, `itemsd_B`, `itemsd_s3`, `itemsd_tbv`, `itemsd_ltex`, `itemsd_nbr`	standard deviation of the score over the 1 024 tokens of the image
image z-score	1	`itemz_q`	$(r_j - \bar{r})/(\text{sd}(r) + 10^{-6})$ within the image
Total	35		

There are 35 features in seven groups; names, order, and counts as in runs/P2-LRF-FROZEN-LFIX/spec.json (features). B_j and D_j use the notation of Section III of the main paper. Source of the formulas: analysis_posthoc/learned_ceiling/data.py, functions feature_names and engineer.

S-V. Symbol Crosswalk

S-VI. Study 2: Reference Encoders and Pixel Parity

The reference encoders are openai/clip-vit-large-patch14 at revision `32bd64288804d66eefd0ccbe215aa642df71cc41` and google/siglip-so400m-patch14-384 at revision `9fdffc58afc957d1a03a25b10dba0329ab15c2a3`, scored with the CLIP cosine score and the SigLIP pairwise logit; the decision is the score of the true caption above that of the wrong caption, with ties counted as wrong. They are run on the same first 450 fold items (300 for the three severity-3 cells) on images regenerated with the runner's corruption functions and seeds; no caption exceeds either text encoder's length limit. Pixel parity: the UniDisc tokeniser's re-encoding of the regenerated images agreed with the stored codes on at least 0.989 of tokens for the clean, JPEG, noise, and splice conditions and at least 0.986 for the clean and severity-3

TABLE S25
CROSSWALK FROM PREREGISTRATION AND CODE NAMES TO THE SYMBOLS OF THE PAPER

Preregistration or code	Paper
Study 1	
`alpha`, `beta` in the signal-to-noise calculator	ϕ_0, ϕ_1
`alpha_obs`, `beta_obs`	$\hat{\phi}_0$, $\hat{\phi}_1$
partition group S_k	group of a token in the K-group partition
$q_j = B_j - D_j$ (statistic `q`)	$-u_j$
GM; I-GATE; I-RAND4	GATE-M; GATE-I; random-gate best-of-four re-infill
Proposition 4 of the method specification	the independent-token condition (Lemma 1 of the main paper)
Proposition 2 of the theory addendum	the correlated-token proposition of the main paper
Proposition 3, Lemma 2, Corollary 3.1 of the theory addendum	Proposition S1, Lemma S1, Corollary S1
Study 2	
`acc_a`, `acc_b` in `a6_analysis.json`	accuracy with LRF, accuracy without weighting
`knots`, `ridge`, `lr`, `T` in `spec.json`	Q, λ, learning rate, T
$L_{\mathrm{FIX}} = 13$	caption-slot length 13
`y_lrf_L`, `y_none_L`	L-LEN decisions with the length-matched wrong caption
feature `q` and its derived names in `spec.json`	$r_j = B_j - D_j$ (Study 2's own symbol; Study 1's q_j is not used in Study 2)
pixel variance in `data.py`	v_j
`sigmoid` in `spec.json`	sigm, the logistic function of the weight model

conditions (five items each; threshold 0.98). These two runs supersede the field `pixel_parity: not run` of the main reference-encoder file, which records only that the main run was made without the parity option; the parity files were moved unchanged from a wrong output directory, and audit-time hashes are recorded. Agreement below one is expected from process-to-process tokenisation differences (S-I, Study 2 deviation 1); the control detects gross pixel errors, not single-pixel ones. No committed run recorded a timing for the encoders, so their cost per item is not reported.

S-VII. STUDY 2: EXPLORATORY ABLATION (TIER 3)

Label. Exploratory; added after the primary, family H, and Study 2 results. No ablation was preregistered. The plan (research/paper2/tnnls/ABLATION-PLAN-2026-10-30.md) was written and then checked by an automated statistical review, and the variant list was committed before any variant was fitted or scored. *Disclosure.* When the plan was written, its author had seen the fresh-fold results of the frozen LRF and of the unweighted rule, an exploratory cross-fit of four learned-weighting families on the design items, and the frozen LRF's shape functions (main paper, Fig. 6); the variant list was fixed after the fresh LRF and unweighted results had been seen, and no variant was fitted or scored before the list was committed. Every listed variant is reported; no variant was selected and scored again, and the tested object of Study 2 stays the frozen LRF.

Variants: All fitted variants use LRF's protocol with the stated change: the same objective, optimiser, nine-configuration search with successive halving, three inner folds over the 500 design items, selection by the inner-fold objective, and one final seed. A-LIN is a linear logit on the 35 features (no hinge terms). A-S-BD, A-S-s3, A-S-tbv, A-S-ltex, A-S-nbr, and A-S-pos are the same additive model on the features computed from one source only: B and D (14 features), s_3 (5), tbv (5), ltex (4), nbr (4), and grid position (3). A-SF is a single-feature weight (feature `rad`, chosen from 35 candidate features by the inner-fold objective on the design items, a selection budget 35 times that of the frozen LRF; the selection used no fresh item). A-MIL is attention-weighted pooling in the style of gated-attention multiple-instance learning (Ilse et al., 2018; cited in the main paper) over the same 35 features, fitted with the same objective, design items, and tuning budget; the evidence Δ_j takes the place of the instance classifier. A-UNL is an unlearned rank weight on s_3, $w_j = 1 - \mathrm{rank}(s_{3,j})$; the direction and the rank transform were fixed by the plan's author, who had seen the frozen LRF's shape functions; they were not read from an accuracy. The knockouts evaluate the frozen LRF with one group set to its design-item mean: by source (KO-S-BD, KO-S-s3, KO-S-tbv, KO-S-ltex, KO-S-nbr, KO-S-pos) and by formula type (KO-F-token, KO-F-position, KO-F-neighbourhood, KO-F-rank, KO-F-item). Setting a group to its mean gives inputs that the model did not see together in fitting; an interval away from zero shows that the frozen model's decisions depend on those inputs, not that the group alone detects anything. KO-S-pos and KO-F-position act on the same three features, so their rows are identical.

Intervals: Each interval is a marginal 95% interval. In all, 336 intervals were computed from the same images (variants × cells × two contrasts); they are positively dependent and no familywise or false-discovery adjustment was made. If every true difference were zero, about 5% of them (about 17) would still exclude zero, and the count of intervals that exclude zero is not a calibrated tally because of the dependence. An interval that excludes zero is therefore a descriptive observation, not a finding; patterns across rows and cells are described, single intervals are not claims. The intervals cover item sampling only: refit-seed variation and design-item sampling are not in them. Intervals are paired percentile bootstrap intervals with 10 000 item resamples, one shared index matrix per cell; differences are variant minus

reference, so a positive value favours the variant, and an interval that contains zero does not show that the difference is zero. The 300 images of the LG cells are the first 300 images of the L1 cells, so the eight cells are not independent samples; the cells have 450 (L1) and 300 (LG) items.

Controls: Recomputed from the stored arrays, the unweighted rule and the frozen LRF reproduced the decisions of the preregistered analysis on all 3 000 item-cells (0 mismatches each), and the knockout of all 35 features (KO-ALL, an exact-equality control with no interval) reproduced the unweighted rule on every item-cell. An LRF refit by the ablation harness at seed 0 decided every fresh item as the frozen LRF did.

Results: Tables S26 and S27 give every variant against the frozen LRF and Table S28 against the unweighted rule. Under Gaussian noise (L1) the intervals against LRF contained zero for A-LIN, A-S-BD, A-S-s3, and A-MIL, and lay below zero for A-S-tbv, A-S-ltex, A-S-nbr, A-S-pos, A-SF, and A-UNL; among knockouts, KO-S-BD and KO-F-neighbourhood lay below zero and KO-S-nbr above zero, and every other knockout interval contained zero. In other cells, intervals against LRF that lay above zero were those of A-LIN under JPEG (0.020 [0.004, 0.038]), A-S-BD under impulse noise (0.043 [0.003, 0.083]), and KO-S-s3 under JPEG (0.011 [0.002, 0.022]); this comparison was added after the Study 2 results and is not a test. The tables give every other cell, row by row, without a summary. The unlearned rank weight on s_3 had a paired difference from the unweighted rule of −0.002 [−0.024, 0.020] and from LRF of −0.071 [−0.111, −0.031] under Gaussian noise. Across the three refit seeds, the range of fresh-fold accuracy was at most 0.030 in every variant and cell (Table S29). Seed 0 is the pre-fixed seed that LRF used, not the best of three; the extra seeds repeat only the final fit, not the configuration search or the choice of design items, so they understate refit variation. The seed-spread rows are refits, not the frozen LRF.

TABLE S26
EXPLORATORY ABLATION (ADDED AFTER THE PRIMARY, FAMILY H, AND STUDY 2 RESULTS): ACCURACY AND DIFFERENCE FROM THE FROZEN LRF, L1 CELLS

	clean ($n = 450$)		JPEG 10 ($n = 450$)		noise 0.1 ($n = 450$)		splice 128 ($n = 450$)	
Row	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]
NONE (reference)	0.887	–	0.849	–	0.733	–	0.896	–
LRF (reference)	0.918	–	0.864	–	0.802	–	0.927	–
A-LIN	0.920	0.002 [−0.013, 0.018]	0.884	0.020 [0.004, 0.038]	0.791	−0.011 [−0.031, 0.009]	0.931	0.004 [−0.009, 0.018]
A-S-BD	0.922	0.004 [−0.016, 0.024]	0.871	0.007 [−0.018, 0.031]	0.800	−0.002 [−0.033, 0.029]	0.911	−0.016 [−0.038, 0.004]
A-S-s3	0.893	−0.024 [−0.051, 0.002]	0.853	−0.011 [−0.040, 0.018]	0.769	−0.033 [−0.071, 0.004]	0.889	−0.038 [−0.062, −0.016]
A-S-tbv	0.913	−0.004 [−0.029, 0.020]	0.856	−0.009 [−0.038, 0.020]	0.762	−0.040 [−0.078, −0.002]	0.904	−0.022 [−0.044, 0.000]
A-S-ltex	0.896	−0.022 [−0.049, 0.004]	0.840	−0.024 [−0.056, 0.004]	0.758	−0.044 [−0.084, −0.004]	0.893	−0.033 [−0.058, −0.011]
A-S-nbr	0.900	−0.018 [−0.042, 0.007]	0.873	0.009 [−0.016, 0.033]	0.724	−0.078 [−0.116, −0.040]	0.898	−0.029 [−0.053, −0.007]
A-S-pos	0.902	−0.016 [−0.044, 0.013]	0.864	0.000 [−0.029, 0.029]	0.731	−0.071 [−0.107, −0.036]	0.891	−0.036 [−0.064, −0.007]
A-SF	0.893	−0.024 [−0.053, 0.002]	0.840	−0.024 [−0.056, 0.007]	0.742	−0.060 [−0.096, −0.024]	0.891	−0.036 [−0.064, −0.007]
A-MIL	0.864	−0.053 [−0.082, −0.024]	0.838	−0.027 [−0.056, 0.002]	0.793	−0.009 [−0.044, 0.027]	0.851	−0.076 [−0.102, −0.049]
A-UNL	0.896	−0.022 [−0.044, 0.000]	0.838	−0.027 [−0.058, 0.004]	0.731	−0.071 [−0.111, −0.031]	0.891	−0.036 [−0.060, −0.013]
KO-S-BD	0.909	−0.009 [−0.027, 0.009]	0.836	−0.029 [−0.053, −0.004]	0.749	−0.053 [−0.082, −0.027]	0.907	−0.020 [−0.038, −0.002]
KO-S-s3	0.913	−0.004 [−0.020, 0.011]	0.876	0.011 [0.002, 0.022]	0.811	0.009 [−0.009, 0.027]	0.918	−0.009 [−0.020, 0.000]
KO-S-tbv	0.904	−0.013 [−0.031, 0.004]	0.880	0.016 [−0.002, 0.036]	0.784	−0.018 [−0.042, 0.004]	0.900	−0.027 [−0.044, −0.009]
KO-S-ltex	0.911	−0.007 [−0.022, 0.007]	0.822	−0.042 [−0.062, −0.024]	0.787	−0.016 [−0.036, 0.004]	0.916	−0.011 [−0.027, 0.004]
KO-S-nbr	0.927	0.009 [−0.011, 0.029]	0.862	−0.002 [−0.024, 0.020]	0.842	0.040 [0.013, 0.069]	0.924	−0.002 [−0.022, 0.018]
KO-S-pos	0.913	−0.004 [−0.020, 0.011]	0.869	0.004 [−0.011, 0.020]	0.791	−0.011 [−0.036, 0.013]	0.918	−0.009 [−0.022, 0.002]
KO-F-token	0.909	−0.009 [−0.033, 0.016]	0.849	−0.016 [−0.049, 0.018]	0.820	0.018 [−0.011, 0.047]	0.902	−0.024 [−0.049, −0.002]
KO-F-position	0.913	−0.004 [−0.020, 0.011]	0.869	0.004 [−0.011, 0.020]	0.791	−0.011 [−0.036, 0.013]	0.918	−0.009 [−0.022, 0.002]
KO-F-neighbourhood	0.884	−0.033 [−0.056, −0.011]	0.771	−0.093 [−0.127, −0.062]	0.740	−0.062 [−0.096, −0.031]	0.871	−0.056 [−0.082, −0.031]
KO-F-rank	0.889	−0.029 [−0.056, −0.004]	0.851	−0.013 [−0.040, 0.013]	0.773	−0.029 [−0.064, 0.007]	0.882	−0.044 [−0.069, −0.020]
KO-F-item	0.916	−0.002 [−0.018, 0.011]	0.871	0.007 [−0.011, 0.024]	0.784	−0.018 [−0.047, 0.009]	0.927	0.000 [−0.011, 0.011]

Differences are variant minus reference; positive favours the variant. Each row is one refit seed with the configuration chosen by the design-item search; the intervals (95%, paired percentile bootstrap, 10 000 resamples) cover item sampling only. KO-ALL is a control (decisions identical to the unweighted rule on every item) and has no interval. Nothing is bold; rows in the order of the plan. Source: runs/P2-LRF-ABL-SCORE/ablation_analysis.json; script paper2-tnnls/scripts/tab_ablation.py.

Deviations entry: Tier-3 ablation (exploratory; added after the primary, family H, and Study 2 results): plan research/paper2/tnnls/ABLATION-PLAN-2026-10-30.md committed before any variant was scored on fresh items; variants fitted only on the 500 design items; every variant reported; intervals only.

S-VIII. SOURCES OF THE MAIN PAPER'S FIGURES AND TABLES

Every figure and table of the main paper is computed by a script in paper2-tnnls/scripts/ that reads only committed run files.

- Fig. 1 (conceptual, no data): script paper2-tnnls/scripts/fig1_measurement.py; definitions in Section III.
- Fig. 3 (theory, no data): script paper2-tnnls/scripts/fig2_theory.py; the shade deepens where more curves lie below, and a curve ends where it reaches $\phi_1 = 1$ or has no root.
- Family P table (Table S5): runs/P2-GATE-M4/analysis.json (primary.per_corruption.*, outcome.label); runs/P2-PAPER-BOUNDS/p_exact_bounds.json.

TABLE S27
EXPLORATORY ABLATION: ACCURACY AND DIFFERENCE FROM THE FROZEN LRF, LG CELLS (THE FIRST 300 L1 IMAGES)

	clean ($n = 300$)		shot ($n = 300$)		impulse ($n = 300$)		speckle ($n = 300$)	
Row	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]	Acc.	d vs LRF [95 %]
NONE (reference)	0.890	–	0.673	–	0.700	–	0.697	–
LRF (reference)	0.913	–	0.767	–	0.750	–	0.757	–
A-LIN	0.917	0.003 [−0.017, 0.023]	0.737	−0.030 [−0.057, −0.007]	0.753	0.003 [−0.023, 0.033]	0.753	−0.003 [−0.033, 0.027]
A-S-BD	0.920	0.007 [−0.017, 0.033]	0.773	0.007 [−0.030, 0.043]	0.793	0.043 [0.003, 0.083]	0.773	0.017 [−0.023, 0.057]
A-S-s3	0.893	−0.020 [−0.050, 0.010]	0.750	−0.017 [−0.060, 0.023]	0.743	−0.007 [−0.053, 0.037]	0.757	0.000 [−0.043, 0.043]
A-S-tbv	0.900	−0.013 [−0.043, 0.017]	0.730	−0.037 [−0.083, 0.007]	0.747	−0.003 [−0.050, 0.043]	0.753	−0.003 [−0.043, 0.033]
A-S-ltex	0.893	−0.020 [−0.053, 0.010]	0.673	−0.093 [−0.143, −0.047]	0.713	−0.037 [−0.087, 0.013]	0.683	−0.073 [−0.117, −0.030]
A-S-nbr	0.893	−0.020 [−0.050, 0.010]	0.710	−0.057 [−0.107, −0.010]	0.690	−0.060 [−0.110, −0.010]	0.720	−0.037 [−0.080, 0.007]
A-S-pos	0.900	−0.013 [−0.047, 0.020]	0.700	−0.067 [−0.110, −0.023]	0.737	−0.013 [−0.060, 0.033]	0.727	−0.030 [−0.070, 0.010]
A-SF	0.883	−0.030 [−0.063, 0.003]	0.700	−0.067 [−0.113, −0.020]	0.727	−0.023 [−0.073, 0.027]	0.730	−0.027 [−0.067, 0.010]
A-MIL	0.857	−0.057 [−0.093, −0.023]	0.760	−0.007 [−0.050, 0.037]	0.763	0.013 [−0.030, 0.057]	0.780	0.023 [−0.023, 0.070]
A-UNL	0.890	−0.023 [−0.050, 0.003]	0.673	−0.093 [−0.143, −0.047]	0.700	−0.050 [−0.100, 0.000]	0.697	−0.060 [−0.103, −0.020]
KO-S-BD	0.903	−0.010 [−0.033, 0.013]	0.703	−0.063 [−0.100, −0.027]	0.720	−0.030 [−0.067, 0.007]	0.720	−0.037 [−0.067, −0.007]
KO-S-s3	0.917	0.003 [−0.013, 0.020]	0.760	−0.007 [−0.033, 0.017]	0.760	0.010 [−0.013, 0.033]	0.777	0.020 [−0.003, 0.043]
KO-S-tbv	0.903	−0.010 [−0.027, 0.007]	0.730	−0.037 [−0.067, −0.007]	0.750	0.000 [−0.030, 0.030]	0.740	−0.017 [−0.047, 0.010]
KO-S-ltex	0.907	−0.007 [−0.027, 0.010]	0.737	−0.030 [−0.057, −0.007]	0.737	−0.013 [−0.033, 0.007]	0.767	0.010 [−0.010, 0.033]
KO-S-nbr	0.917	0.003 [−0.020, 0.027]	0.790	0.023 [−0.010, 0.060]	0.783	0.033 [−0.007, 0.073]	0.783	0.027 [−0.013, 0.067]
KO-S-pos	0.913	0.000 [−0.017, 0.017]	0.763	−0.003 [−0.030, 0.023]	0.750	0.000 [−0.027, 0.027]	0.750	−0.007 [−0.030, 0.017]
KO-F-token	0.900	−0.013 [−0.043, 0.017]	0.773	0.007 [−0.027, 0.040]	0.770	0.020 [−0.017, 0.057]	0.770	0.013 [−0.020, 0.047]
KO-F-position	0.913	0.000 [−0.017, 0.017]	0.763	−0.003 [−0.030, 0.023]	0.750	0.000 [−0.027, 0.027]	0.750	−0.007 [−0.030, 0.017]
KO-F-neighbourhood	0.863	−0.050 [−0.080, −0.020]	0.697	−0.070 [−0.110, −0.033]	0.717	−0.033 [−0.077, 0.007]	0.717	−0.040 [−0.077, −0.003]
KO-F-rank	0.877	−0.037 [−0.067, −0.007]	0.727	−0.040 [−0.080, 0.000]	0.753	0.003 [−0.043, 0.050]	0.723	−0.033 [−0.073, 0.007]
KO-F-item	0.917	0.003 [−0.013, 0.020]	0.733	−0.033 [−0.070, 0.000]	0.747	−0.003 [−0.040, 0.033]	0.737	−0.020 [−0.050, 0.010]

As Table S26. The LG cells use the first 300 images of the L1 cells, so the cells are not independent samples.

TABLE S28
EXPLORATORY ABLATION: DIFFERENCE FROM THE UNWEIGHTED RULE [95% INTERVAL], ALL CELLS

	L1 cells ($n = 450$)				LG cells ($n = 300$)			
Row	clean	JPEG 10	noise 0.1	splice 128	clean	shot	impulse	speckle
A-LIN	0.033 [0.009, 0.060]	0.036 [0.007, 0.064]	0.058 [0.020, 0.096]	0.036 [0.011, 0.062]	0.027 [0.000, 0.057]	0.063 [0.017, 0.110]	0.053 [0.000, 0.107]	0.057 [0.013, 0.100]
A-S-BD	0.036 [0.013, 0.058]	0.022 [0.000, 0.044]	0.067 [0.033, 0.100]	0.016 [−0.009, 0.040]	0.030 [0.007, 0.057]	0.100 [0.057, 0.147]	0.093 [0.053, 0.137]	0.077 [0.033, 0.120]
A-S-s3	0.007 [−0.011, 0.024]	0.004 [−0.016, 0.024]	0.036 [0.007, 0.064]	−0.007 [−0.024, 0.011]	0.003 [−0.017, 0.023]	0.077 [0.040, 0.113]	0.043 [0.013, 0.077]	0.060 [0.023, 0.097]
A-S-tbv	0.027 [0.004, 0.049]	0.007 [−0.020, 0.033]	0.029 [0.007, 0.053]	0.009 [−0.013, 0.031]	0.010 [−0.013, 0.037]	0.057 [0.027, 0.090]	0.047 [0.020, 0.077]	0.057 [0.020, 0.093]
A-S-ltex	0.009 [−0.016, 0.033]	−0.009 [−0.038, 0.018]	0.024 [−0.002, 0.053]	−0.002 [−0.024, 0.020]	0.003 [−0.027, 0.030]	0.000 [−0.010, 0.010]	0.013 [−0.010, 0.040]	−0.013 [−0.033, 0.003]
A-S-nbr	0.013 [−0.007, 0.033]	0.024 [0.000, 0.049]	−0.009 [−0.033, 0.016]	0.002 [−0.020, 0.024]	0.003 [−0.020, 0.030]	0.037 [0.007, 0.070]	−0.010 [−0.043, 0.023]	0.023 [−0.007, 0.057]
A-S-pos	0.016 [−0.013, 0.044]	0.016 [−0.013, 0.047]	−0.002 [−0.036, 0.031]	−0.004 [−0.036, 0.024]	0.010 [−0.023, 0.043]	0.027 [−0.010, 0.067]	0.037 [−0.003, 0.077]	0.030 [−0.007, 0.067]
A-SF	0.007 [−0.022, 0.036]	−0.009 [−0.040, 0.022]	0.009 [−0.024, 0.042]	−0.004 [−0.036, 0.027]	−0.007 [−0.040, 0.027]	0.027 [−0.013, 0.067]	0.027 [−0.017, 0.070]	0.033 [−0.007, 0.073]
A-MIL	−0.022 [−0.056, 0.011]	−0.011 [−0.049, 0.027]	0.060 [0.018, 0.104]	−0.044 [−0.080, −0.009]	−0.033 [−0.073, 0.007]	0.087 [0.030, 0.143]	0.063 [0.007, 0.120]	0.083 [0.030, 0.137]
A-UNL	0.009 [−0.011, 0.029]	−0.011 [−0.036, 0.011]	−0.002 [−0.024, 0.020]	−0.004 [−0.024, 0.016]	0.000 [−0.023, 0.023]	0.000 [−0.023, 0.023]	0.000 [−0.030, 0.030]	0.000 [−0.030, 0.030]
KO-S-BD	0.022 [−0.007, 0.051]	−0.013 [−0.044, 0.018]	0.016 [−0.018, 0.051]	0.011 [−0.018, 0.040]	0.013 [−0.020, 0.047]	0.030 [−0.010, 0.073]	0.020 [−0.023, 0.063]	0.023 [−0.017, 0.060]
KO-S-s3	0.027 [0.002, 0.053]	0.027 [−0.002, 0.056]	0.078 [0.040, 0.118]	0.022 [−0.002, 0.047]	0.027 [0.000, 0.057]	0.087 [0.040, 0.133]	0.060 [0.013, 0.107]	0.080 [0.037, 0.123]
KO-S-tbv	0.018 [−0.009, 0.044]	0.031 [0.002, 0.060]	0.051 [0.013, 0.089]	0.004 [−0.024, 0.033]	0.013 [−0.017, 0.047]	0.057 [0.013, 0.103]	0.050 [0.003, 0.097]	0.043 [0.000, 0.087]
KO-S-ltex	0.024 [−0.002, 0.051]	−0.027 [−0.060, 0.007]	0.053 [0.016, 0.091]	0.020 [−0.007, 0.049]	0.017 [−0.017, 0.050]	0.063 [0.017, 0.110]	0.037 [−0.007, 0.080]	0.070 [0.023, 0.117]
KO-S-nbr	0.040 [0.013, 0.067]	0.013 [−0.016, 0.042]	0.109 [0.071, 0.149]	0.029 [0.002, 0.056]	0.027 [−0.003, 0.060]	0.117 [0.067, 0.167]	0.083 [0.033, 0.133]	0.087 [0.037, 0.137]
KO-S-pos	0.027 [0.002, 0.051]	0.020 [−0.009, 0.047]	0.058 [0.022, 0.096]	0.022 [−0.002, 0.047]	0.023 [−0.003, 0.053]	0.090 [0.043, 0.140]	0.050 [0.007, 0.093]	0.053 [0.013, 0.093]
KO-F-token	0.022 [−0.007, 0.051]	0.000 [−0.029, 0.029]	0.087 [0.051, 0.122]	0.007 [−0.022, 0.036]	0.010 [−0.023, 0.043]	0.100 [0.057, 0.143]	0.070 [0.027, 0.113]	0.073 [0.030, 0.117]
KO-F-position	0.027 [0.002, 0.051]	0.020 [−0.009, 0.047]	0.058 [0.022, 0.096]	0.022 [−0.002, 0.047]	0.023 [−0.003, 0.053]	0.090 [0.043, 0.140]	0.050 [0.007, 0.093]	0.053 [0.013, 0.093]
KO-F-neighbourhood	−0.002 [−0.036, 0.031]	−0.078 [−0.118, −0.040]	0.007 [−0.031, 0.044]	−0.024 [−0.058, 0.007]	−0.027 [−0.067, 0.013]	0.023 [−0.023, 0.070]	0.017 [−0.033, 0.067]	0.020 [−0.027, 0.067]
KO-F-rank	0.002 [−0.027, 0.029]	0.002 [−0.029, 0.033]	0.040 [0.002, 0.078]	−0.013 [−0.042, 0.016]	−0.013 [−0.047, 0.020]	0.053 [0.010, 0.100]	0.053 [0.007, 0.100]	0.027 [−0.020, 0.073]
KO-F-item	0.029 [0.004, 0.053]	0.022 [−0.007, 0.051]	0.051 [0.020, 0.082]	0.031 [0.007, 0.056]	0.027 [0.000, 0.057]	0.060 [0.023, 0.100]	0.047 [0.013, 0.080]	0.040 [0.003, 0.077]

Variant minus the unweighted rule; positive favours the variant. The interval of LRF against the unweighted rule is the preregistered one (main paper, Study 2 results table) and is not recomputed here.

TABLE S29
EXPLORATORY ABLATION: SEED-SPREAD REFITS (ACCURACY AT SEEDS 0, 1, 2; MEAN, SD, RANGE)

	L1 cells ($n = 450$)				LG cells ($n = 300$)			
Refit	clean	JPEG 10	noise 0.1	splice 128	clean	shot	impulse	speckle
LRF refit (harness)	0.918, 0.900, 0.913	0.864, 0.844, 0.838	0.802, 0.787, 0.780	0.927, 0.904, 0.916	0.913, 0.893, 0.910	0.767, 0.747, 0.737	0.750, 0.757, 0.760	0.757, 0.760, 0.760
	0.910, 0.009, 0.018	0.849, 0.014, 0.027	0.790, 0.011, 0.022	0.916, 0.011, 0.022	0.906, 0.011, 0.020	0.750, 0.015, 0.030	0.756, 0.005, 0.010	0.759, 0.002, 0.003
A-LIN	0.920, 0.920, 0.920	0.884, 0.869, 0.869	0.791, 0.787, 0.782	0.931, 0.927, 0.924	0.917, 0.913, 0.910	0.737, 0.743, 0.747	0.753, 0.757, 0.747	0.753, 0.753, 0.757
	0.920, 0.000, 0.000	0.874, 0.009, 0.016	0.787, 0.004, 0.009	0.927, 0.003, 0.007	0.913, 0.003, 0.007	0.742, 0.005, 0.010	0.752, 0.005, 0.010	0.754, 0.002, 0.003
A-S-BD	0.922, 0.920, 0.920	0.871, 0.873, 0.871	0.800, 0.802, 0.802	0.911, 0.916, 0.913	0.920, 0.913, 0.913	0.773, 0.793, 0.787	0.793, 0.797, 0.793	0.773, 0.783, 0.777
	0.921, 0.001, 0.002	0.872, 0.001, 0.002	0.801, 0.001, 0.002	0.913, 0.002, 0.004	0.916, 0.004, 0.007	0.784, 0.010, 0.020	0.794, 0.002, 0.003	0.778, 0.005, 0.010
A-S-s3	0.893, 0.896, 0.896	0.853, 0.847, 0.851	0.769, 0.776, 0.778	0.889, 0.889, 0.887	0.893, 0.900, 0.897	0.750, 0.747, 0.750	0.743, 0.743, 0.747	0.757, 0.757, 0.757
	0.895, 0.001, 0.002	0.850, 0.003, 0.007	0.774, 0.005, 0.009	0.888, 0.001, 0.002	0.897, 0.003, 0.007	0.749, 0.002, 0.003	0.744, 0.002, 0.003	0.757, 0.000, 0.000
A-S-tbv	0.913, 0.907, 0.909	0.856, 0.844, 0.844	0.762, 0.767, 0.762	0.904, 0.913, 0.907	0.900, 0.910, 0.907	0.730, 0.730, 0.720	0.747, 0.753, 0.753	0.753, 0.773, 0.763
	0.910, 0.003, 0.007	0.848, 0.006, 0.011	0.764, 0.003, 0.004	0.908, 0.005, 0.009	0.906, 0.005, 0.010	0.727, 0.006, 0.010	0.751, 0.004, 0.007	0.763, 0.010, 0.020
A-S-ltex	0.896, 0.893, 0.893	0.840, 0.838, 0.838	0.758, 0.751, 0.753	0.893, 0.891, 0.896	0.893, 0.890, 0.887	0.673, 0.673, 0.667	0.713, 0.723, 0.723	0.683, 0.683, 0.683
	0.894, 0.001, 0.002	0.839, 0.001, 0.002	0.754, 0.003, 0.007	0.893, 0.002, 0.004	0.890, 0.003, 0.007	0.671, 0.004, 0.007	0.720, 0.006, 0.010	0.683, 0.000, 0.000
A-S-nbr	0.900, 0.902, 0.902	0.873, 0.876, 0.876	0.724, 0.724, 0.722	0.898, 0.900, 0.900	0.893, 0.900, 0.897	0.710, 0.710, 0.710	0.690, 0.693, 0.693	0.720, 0.723, 0.723
	0.901, 0.001, 0.002	0.875, 0.001, 0.002	0.724, 0.001, 0.002	0.899, 0.001, 0.002	0.897, 0.003, 0.007	0.710, 0.000, 0.000	0.692, 0.002, 0.003	0.722, 0.002, 0.003
A-S-pos	0.902, 0.907, 0.896	0.864, 0.869, 0.858	0.731, 0.731, 0.742	0.891, 0.893, 0.889	0.900, 0.900, 0.887	0.700, 0.700, 0.707	0.737, 0.740, 0.737	0.727, 0.727, 0.727
	0.901, 0.006, 0.011	0.864, 0.006, 0.011	0.735, 0.006, 0.011	0.891, 0.002, 0.004	0.896, 0.008, 0.013	0.702, 0.004, 0.007	0.738, 0.002, 0.003	0.727, 0.000, 0.000
A-SF	0.893, 0.893, 0.896	0.840, 0.840, 0.836	0.742, 0.740, 0.744	0.891, 0.891, 0.889	0.883, 0.883, 0.887	0.700, 0.700, 0.700	0.727, 0.727, 0.727	0.730, 0.730, 0.730
	0.894, 0.001, 0.002	0.839, 0.003, 0.004	0.742, 0.002, 0.004	0.890, 0.001, 0.002	0.884, 0.002, 0.003	0.700, 0.000, 0.000	0.727, 0.000, 0.000	0.730, 0.000, 0.000
A-MIL	0.864, 0.860, 0.864	0.838, 0.833, 0.838	0.793, 0.807, 0.809	0.851, 0.853, 0.869	0.857, 0.863, 0.857	0.760, 0.760, 0.770	0.763, 0.773, 0.777	0.780, 0.767, 0.777
	0.863, 0.003, 0.004	0.836, 0.003, 0.004	0.803, 0.008, 0.016	0.858, 0.010, 0.018	0.859, 0.004, 0.007	0.763, 0.006, 0.010	0.771, 0.007, 0.013	0.774, 0.007, 0.013

Seed 0 is the pre-fixed seed, not the best of three; the extra seeds repeat only the final fit with the configuration of the seed-0 search, so they understate refit variation. "LRF refit (harness)" is a seed-spread refit, not the frozen LRF. Descriptive; no interval.

- Fig. 4 (family P): runs/P2-GATE-M4/analysis.json, keys primary.per_corruption.*; script paper2-tnnls/scripts/fig3_study1_P.py.
- Table S10 (positive control): runs/P2-GATE-M4/analysis.json (ps.per_corruption.*, ps.positive_control); runs/P2-A23-AUDIT/a23_ps_audit.json.
- Fig. 2 (gate method and calibration check): runs/P2-GATE-M4/analysis.json, keys calibration.fap_check.per_alpha.*, calibration.fap_check.critical_value, calibration.split_replication.*; runs/P2-GATE-M2/threshold.json (n_cal); plotted values in runs/P2-PAPER-FIGS/fig_gate_method.json; script paper2-tnnls/scripts/fig_gate_method.py.
- Fig. 5 (token detection; values in Table S11): runs/P2-GATE-M1/analysis.json, keys table.*, paired_q_vs_best_cheap.*; runs/P2-GATE-M4/analysis.json, keys s_d2.per_corruption.*; plotted values in runs/P2-PAPER-FIGS/fig_detection.json; script paper2-tnnls/scripts/fig_detection.py.
- Feature groups (Fig. 6(a) of the main paper): runs/P2-LRF-FROZEN-LFIX/spec.json (features); analysis_posthoc/learned_ceiling/data.py, functions feature_names and engineer; full table in Table S24.
- Fig. 6 (frozen weight model): runs/P2-LRF-FROZEN-LFIX/{model.npz,spec.json}, runs/P2-LRF-L1/arrays/item000{0,1,2}.npz (SHA-256 checked against arrays_manifest.json); computed values in runs/P2-PAPER-FIGS/; script paper2-tnnls/scripts/fig4_study2_weights.py (also fig4_meta.json). Each curve of panel (b) runs between the lowest and highest of the eight knots of its feature; the effect range is the range of w over the plotted domain.
- Table II (Study 2 families): runs/P2-LRF-L1/a6_analysis.json (family_L.*); runs/P2-LRF-LG/a6_analysis.json (family_LG.cells.*).
- Fig. 7 (family L and L-NI): runs/P2-LRF-L1/a6_analysis.json, keys family_L.*; script paper2-tnnls/scripts/fig5_study2_results.py.
- Figure S1 (item-level margins, descriptive): runs/P2-LRF-L1/predictions.jsonl (fields condition, idx, m_none, m_lrf, y_none, y_lrf; SHA-256 checked against a6_analysis.json); plotted values in runs/P2-PAPER-FIGS/fig_margins.json; script paper2-tnnls/scripts/fig_margins.py.
- Fig. 8 (exploratory ablation): runs/P2-LRF-ABL-SCORE/ablation_analysis.json, keys rows[*].cells.L1_noise0.1.vs_LRF.*, ko_all_control; plotted values in runs/P2-PAPER-FIGS/fig_ablation.json; script paper2-tnnls/scripts/fig_ablation.py; rows in the order of Section S-VII.
- Table III (reference encoders): runs/P2-LRF-BASELINES/baselines.json (table.*); runs/P2-PAPER-TABLES/baselines_clean300.json; revisions in Section S-VI.
  Pixel parity: runs/P2-LRF-BASELINES-PARITY-L1 and runs/P2-LRF-BASELINES-PARITY-LG.